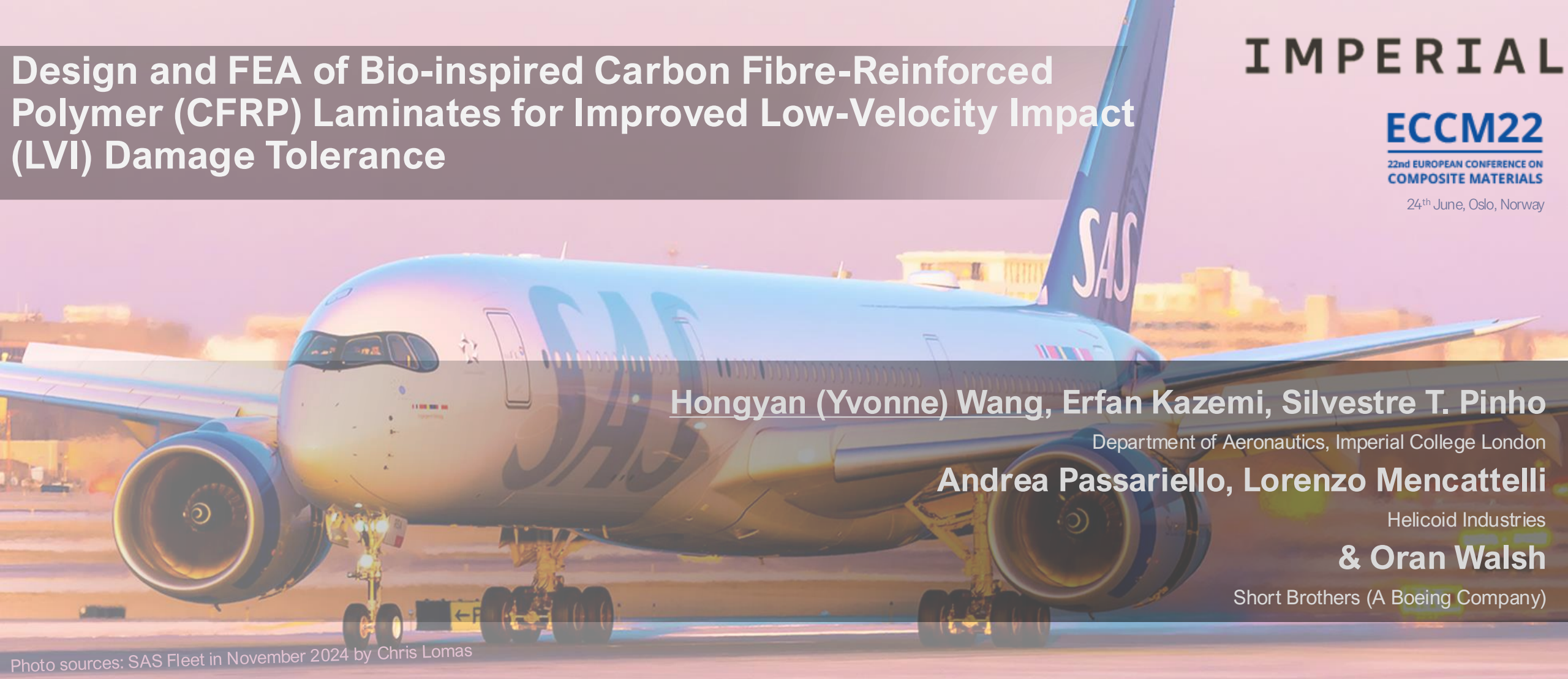
Design and FEA of Bio-inspired Carbon Fibre-Reinforced Polymer (CFRP) Laminates for Improved Low-Velocity Impact (LVI) Damage Tolerance

IMPERIAL

ECCM22
22nd EUROPEAN CONFERENCE ON COMPOSITE MATERIALS
24th June, Oslo, Norway



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Helicoid Industries
& Oran Walsh
Short Brothers (A Boeing Company)

Photo sources: SAS Fleet in November 2024 by Chris Lomas

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https://github.com/Hongyan-Wang
https://pinholab.cc.ic.ac.uk

UKRI-Innovate project - IVI
(Isothermal Variable-volume Infusion)

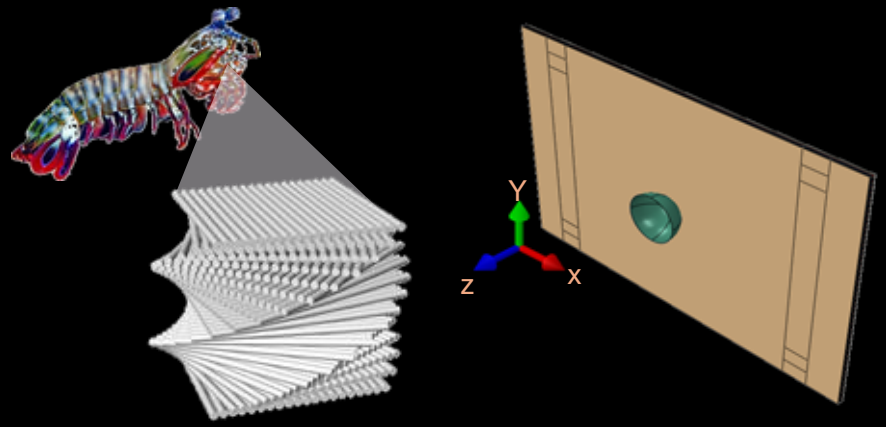
HELICOID INDUSTRIES LTD
Short Brothers A Boeing Company
Department for Business & Trade
AEROSPACE TECHNOLOGY INSTITUTE
UKRI Innovate UK

Outline

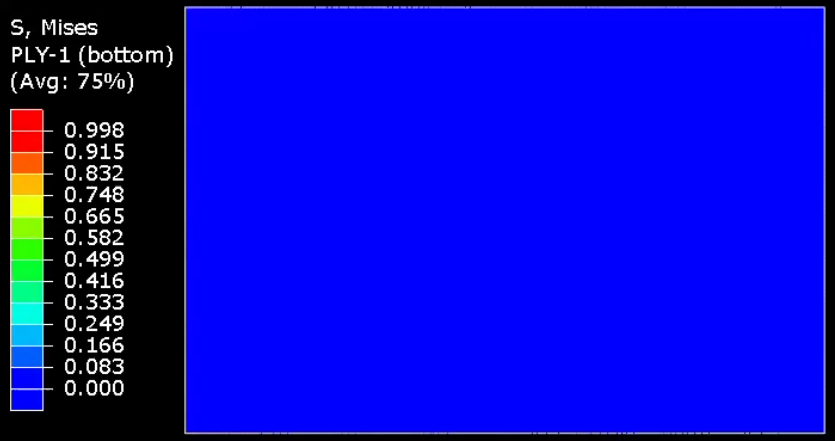
Background & Motivation



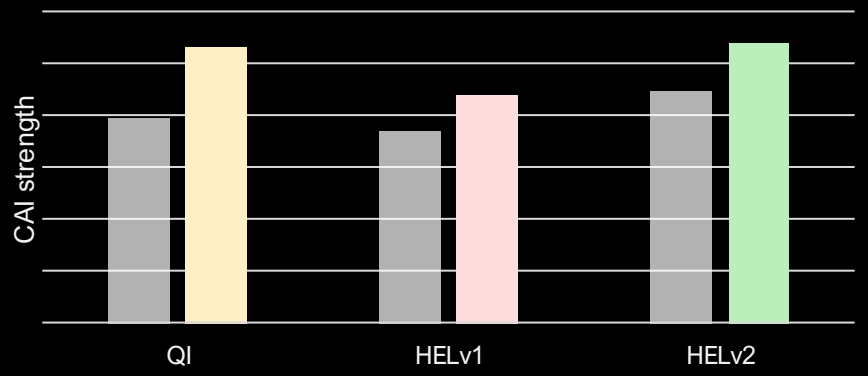
Design & Methodology



Results & Discussion



Summary & Conclusions



Background & Motivation



Photo sources: SAS Scandinavian Airlines Airbus A350-941 by Clément Alloing

Background & Motivation

>50 wt%

Carbon Fibre-Reinforced Polymer (CFRP) composites in A350 and B787

20-22%

Weight reduction achieved

20-25%

Fuel savings achieved [1]



Current challenges

- Complex, multi-axial, and uncertain loading scenarios
- Exposure to harsh environments and impact threats
- Limited damage tolerance in conventional CFRP layups

Proposed approach

Bio-inspired laminate design to enhance impact damage tolerance and maintain structural integrity under critical loading conditions.

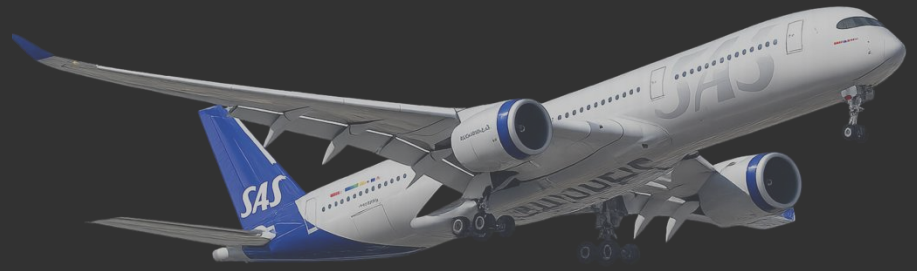
Photo sources: SAS Scandinavian Airlines Airbus A350-941 by Clément Alloing

[1] Zhang, J., et al. (2023). "Past, present and future prospective of global carbon fibre composite developments and applications." Composites Part B: Engineering 250: 110463.

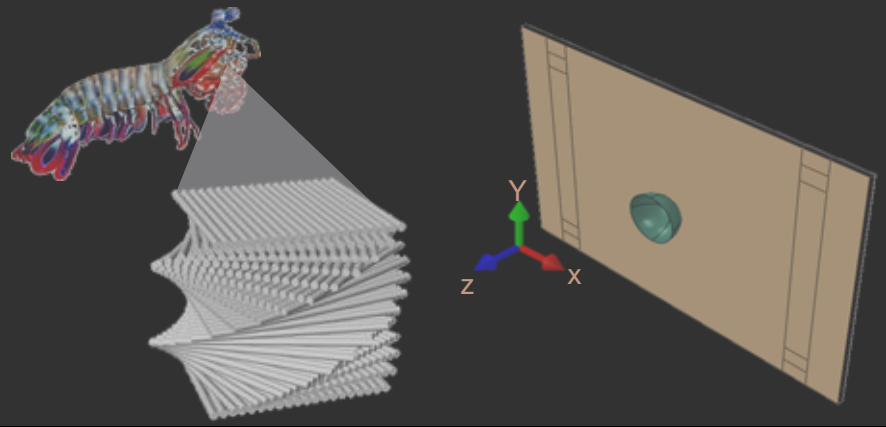
4

Outline

Background & Motivation



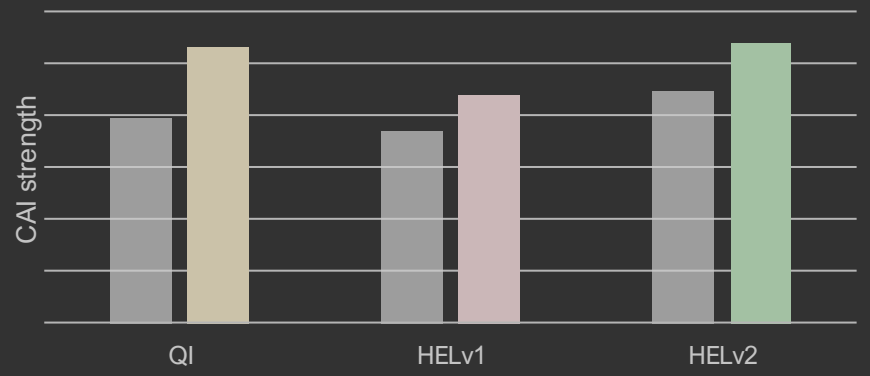
Design & Methodology



Results & Discussion



Summary & Conclusions

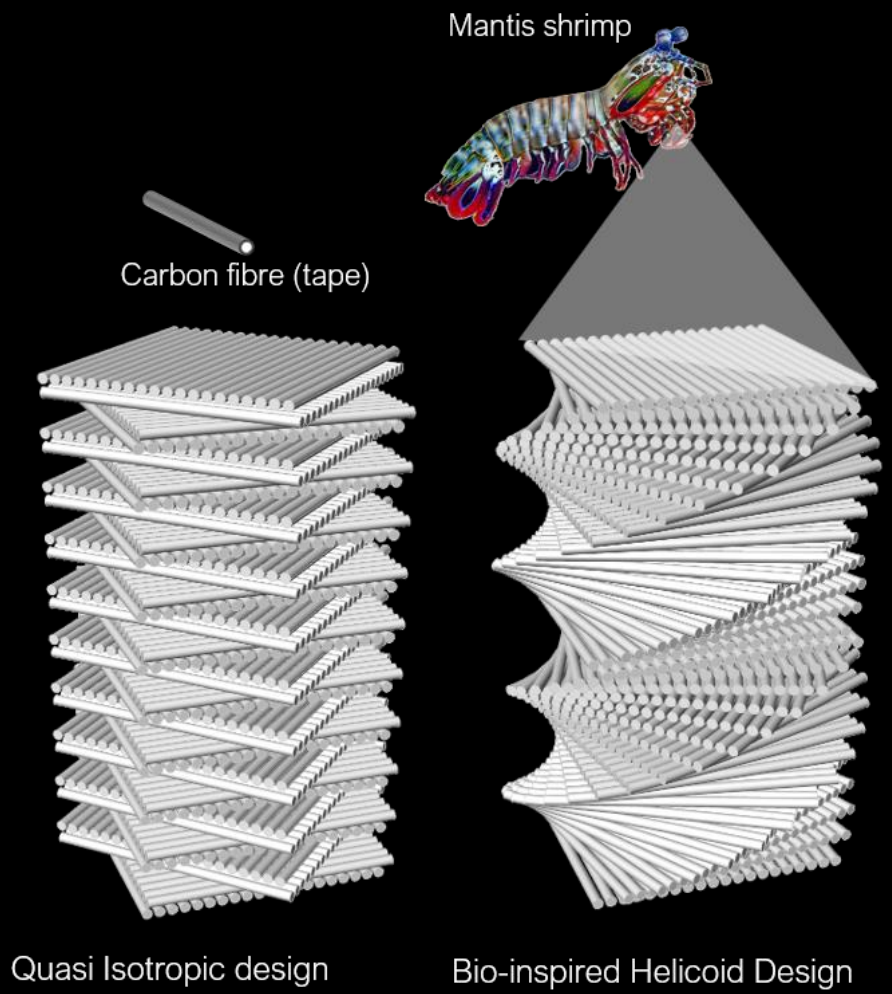


Design & Methodology

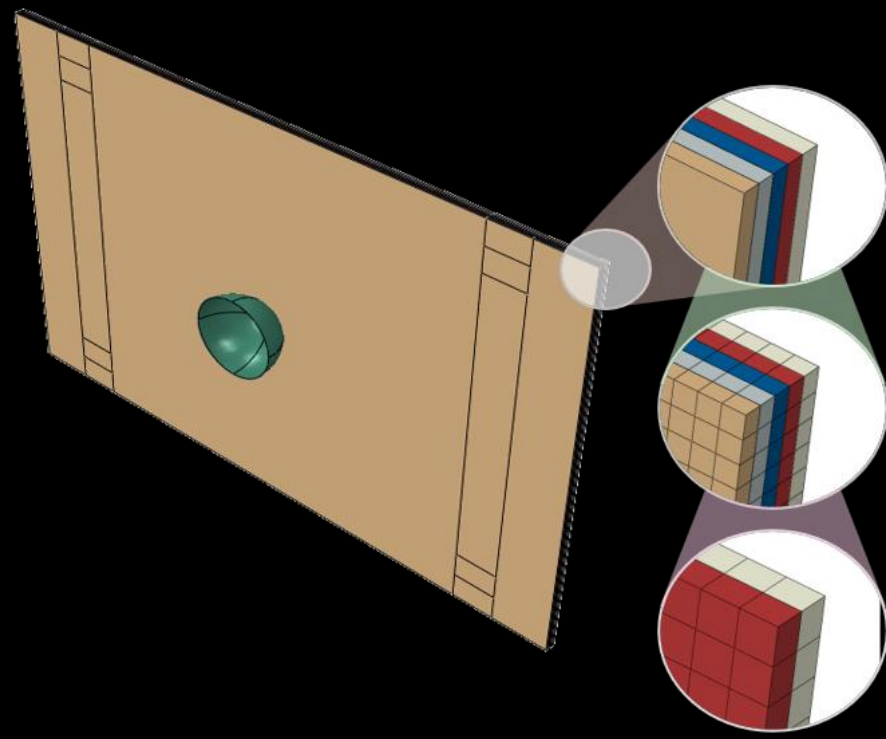
Design concept

Experimental setup

FEA setup





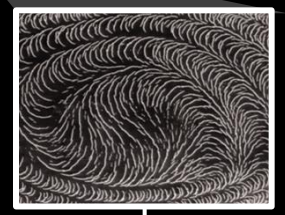


Design & Methodology

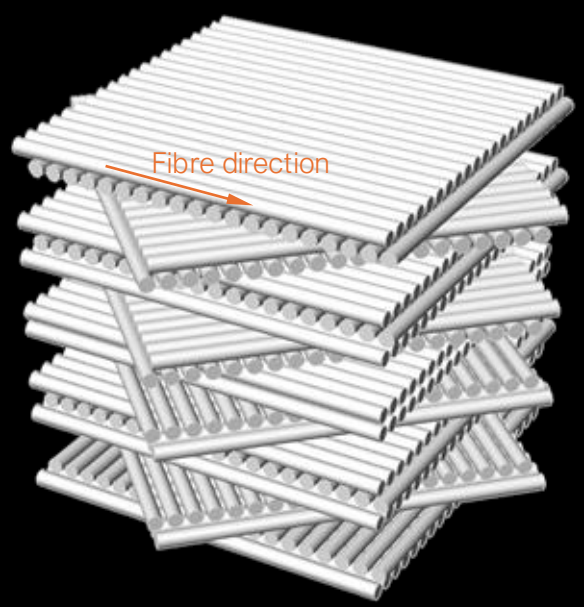
Design concept | Experimental setup | FEA setup

- The baseline layup is a conventional quasi-isotropic (QI) layup
- The helicoidal concept is from the periodic region of the dactyl club of a mantis shrimp [1]
- The proposed helicoidal layups were designed to have comparable D-matrix to QI layup.
- HELv2 has a smaller pitch angle and smoother fibre transition.



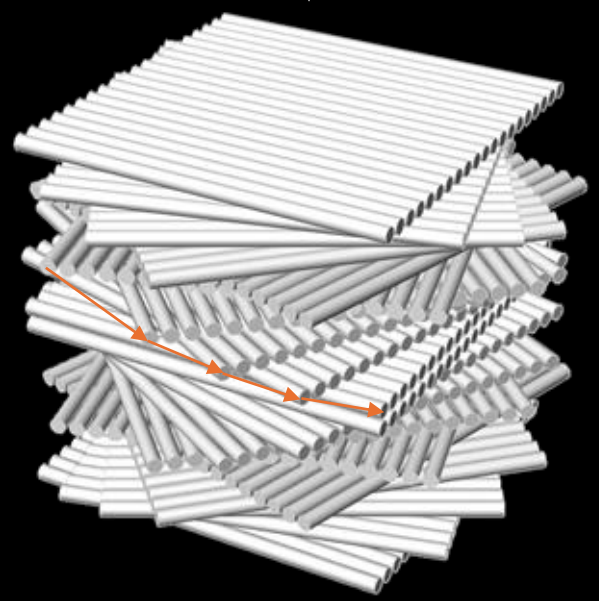


SEM of a damaged coronal cross section [2]



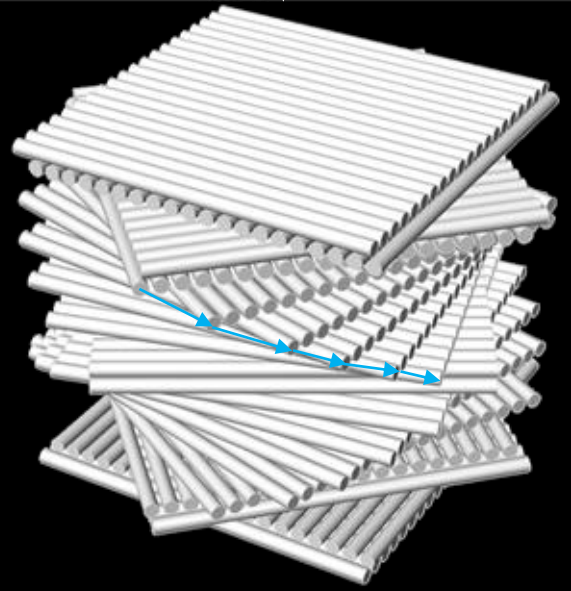
Baseline Quasi-isotropic (QI)

$[0/90/\pm45/0/90/0/\pm45/0]_s$



Helicoid v1 (HELv1)

$[0/15/30/45/75/-75/-45/-30/-15/0]_s$



Helicoid v2 (HELv2)

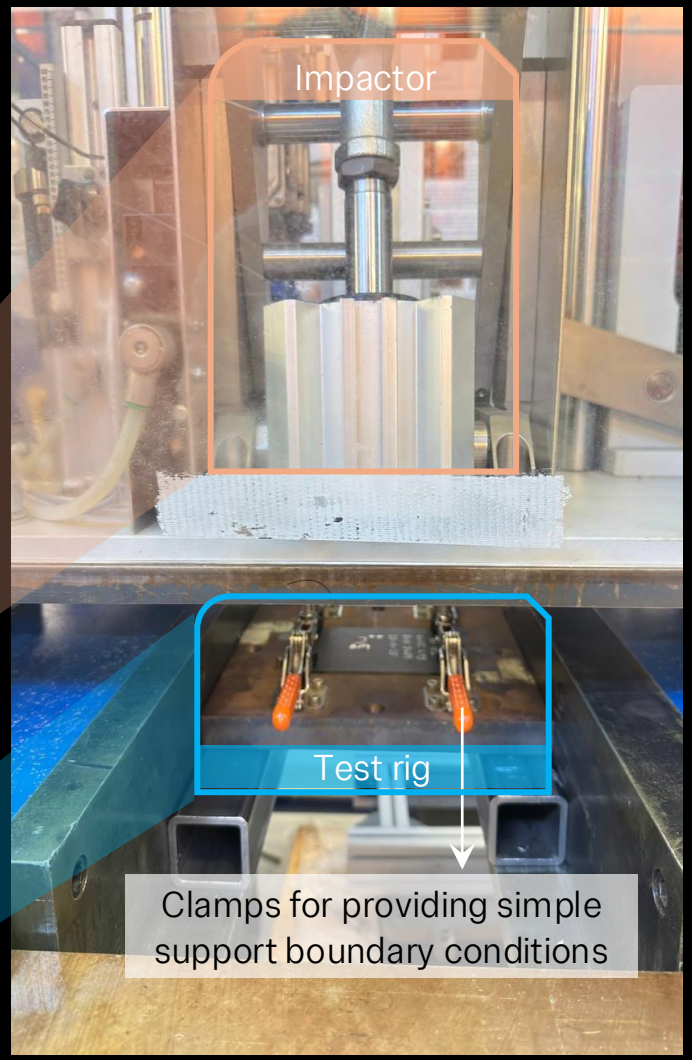
$[0/90/45/-45/-30/-15/-5/5/15/30]_s$

[1] Weaver, J. C., et al. (2012). "The Stomatopod Dactyl Club: A Formidable Damage-Tolerant Biological Hammer." Science 336(6086): 1275-1280.
[2] The ArthroBlogger. (2020, October 13). "Beautiful & terrifying mantis shrimp". Retrieved from https://thearthroblogger.com/2020/10/13/beautiful-terrifying-mantis-shrimp/

Design & Methodology

Design concept | Experimental setup | FEA setup





We tested:

- Low-Velocity Impact (LVI)
- Compression after Impact (CAI)

for the three designs:

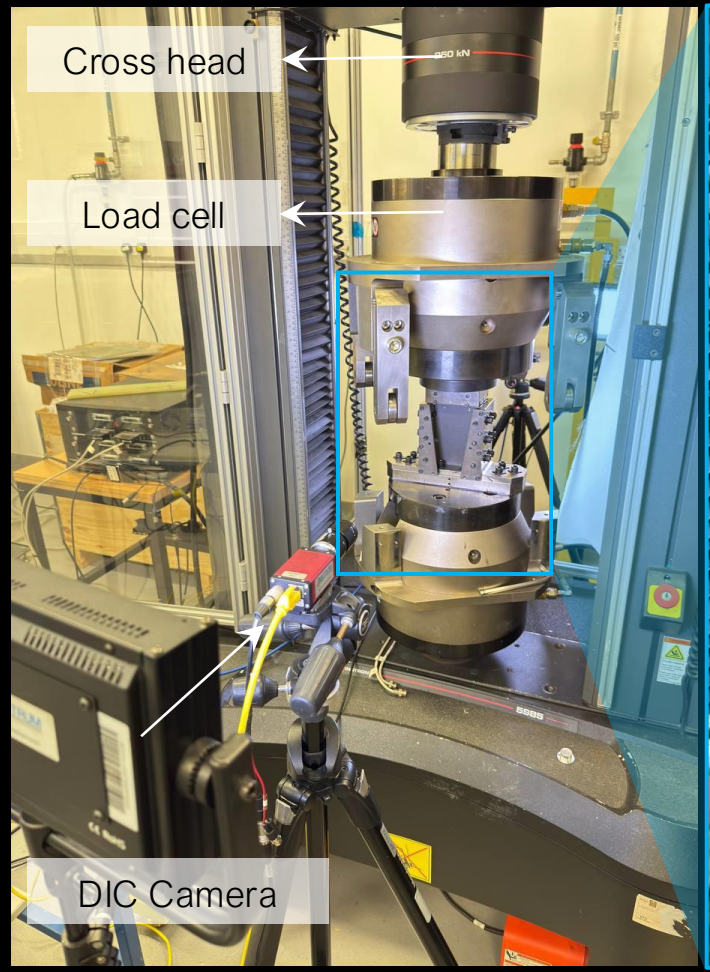
QI, HELv1, HELv2

The specific experimental parameters for LVI are as below:

Test standard	ASTM 7136
Material	IM7/8552
Specimen dimensions	150 mm x 100 mm x 2.3 mm
Impactor diameter	20 mm
Impactor mass	10.5 kg
Impact energy	15 J (initial velocity: 1.69 m/s)
Repeats	3 specimens per layup
Experimentally-measured variables	Time (ms) Force (kN)
Sampling frequency	500 kHz

Design & Methodology

Design concept | Experimental setup | FEA setup





We tested:

- Low-Velocity Impact (LVI)
- Compression after Impact (CAI)

for the three designs:

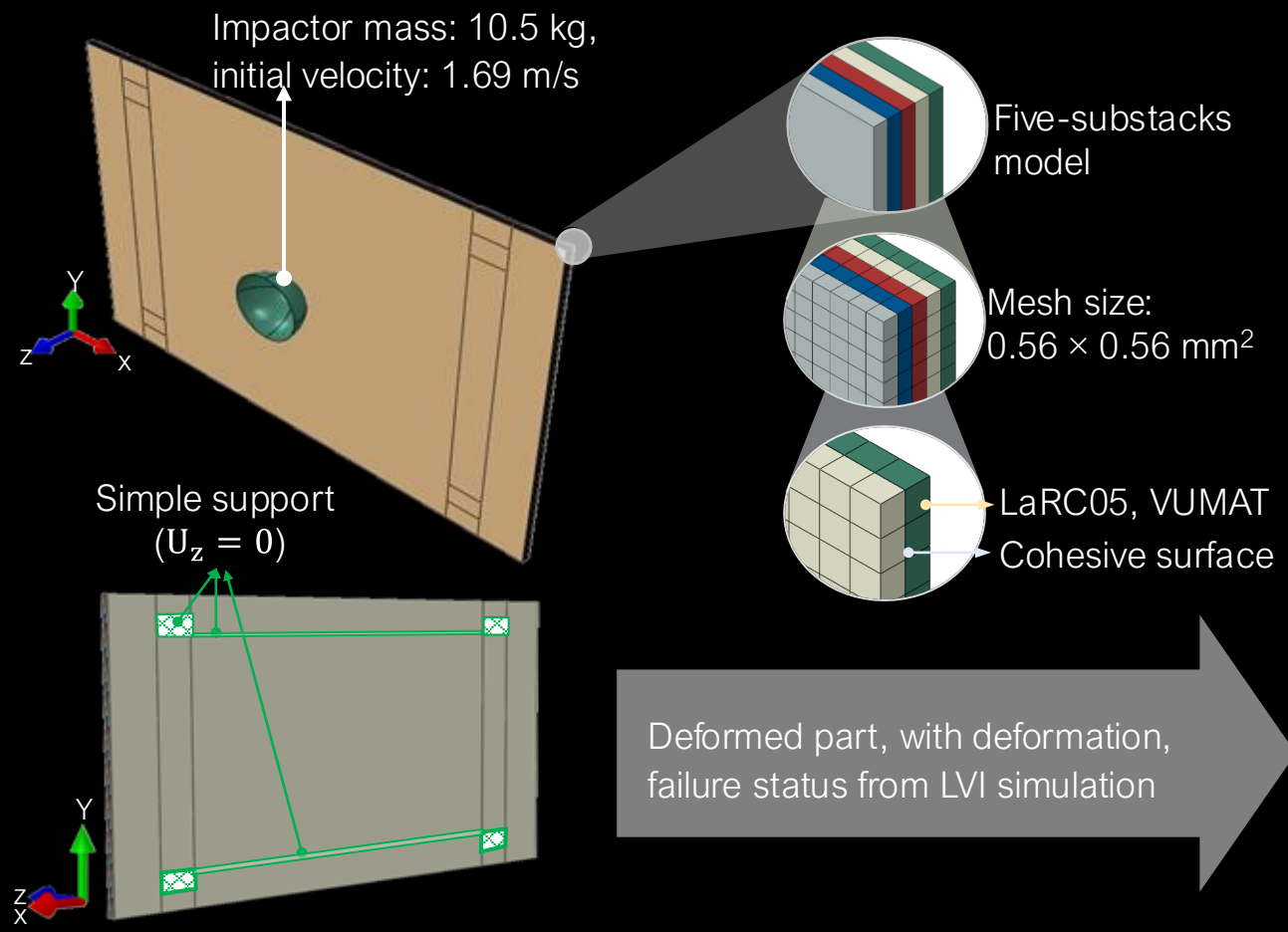
QI, HELv1, HELv2

The specific experimental parameters for CAI are as below:

Test standard	ASTM 7137
Material	IM7/8552
Specimen dimensions	150 mm x 100 mm x 2.3 mm
Repeats	3 impacted specimens per layup
Loading rate	1 mm/min
Experimentally-measured variables	• Crosshead displacement • Force (kN)
DIC cameras	Front and rear cameras for monitoring out-of-plane deformation

Design & Methodology

Design concept | Experimental setup | FEA setup



BCs for LVI analysis

BCs for CAI analysis

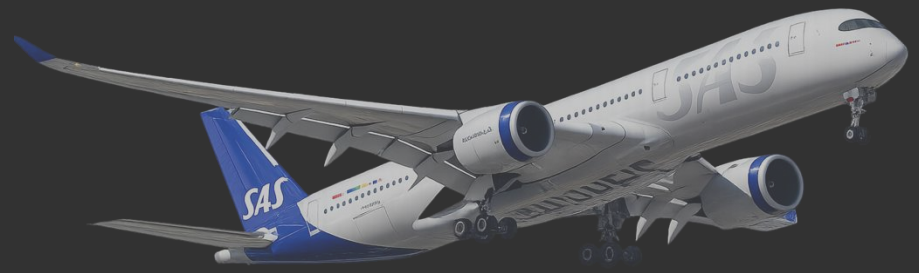
- Abaqus/Explicit simulations
- LaRC05 damage model implemented through a VUMAT subroutine [1]
- LVI and CAI boundary conditions (BCs) follow the respective ASTM standards

- Only the specimen is modelled; testing rigs are excluded

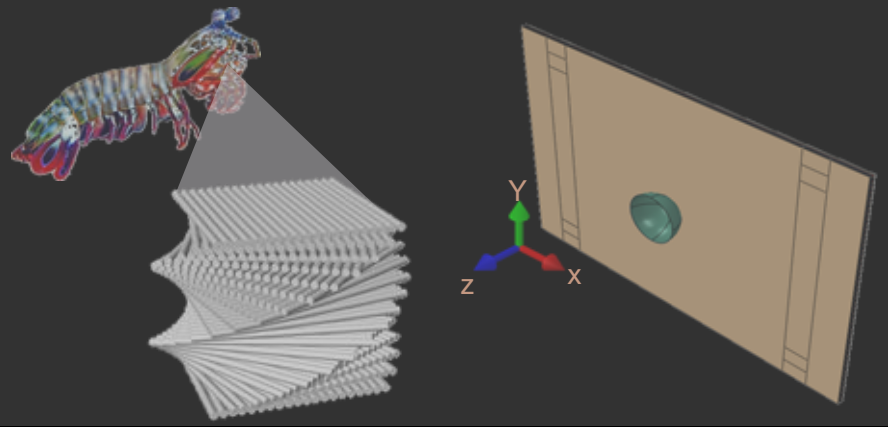
[1] ST Pinho et al. Material and structural response of polymer-matrix fibre-reinforced composites. Journal of Composite Materials. 46, 2313–2341, 2012

Outline

Background & Motivation



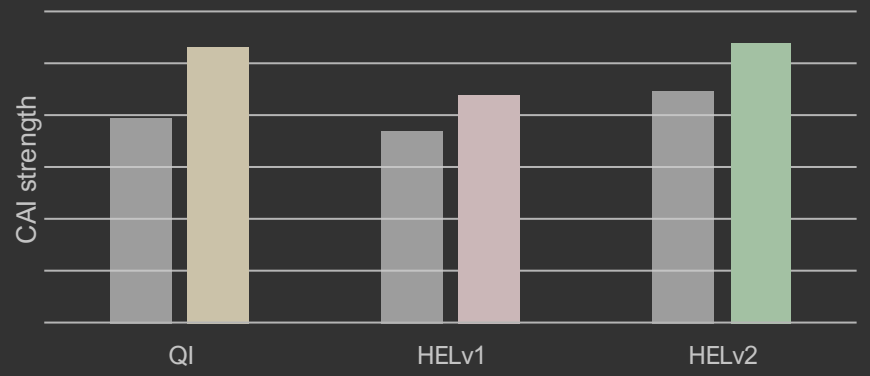
Design & Methodology



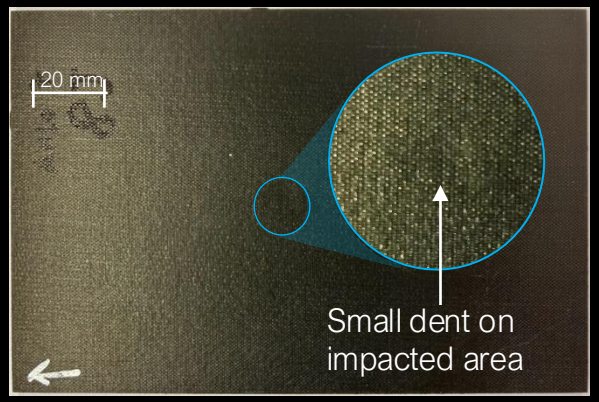
Results & Discussion

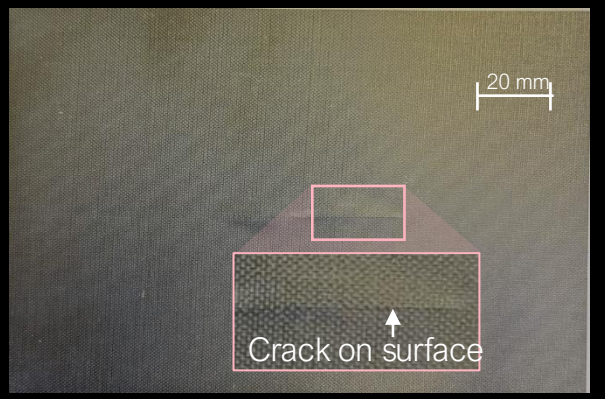


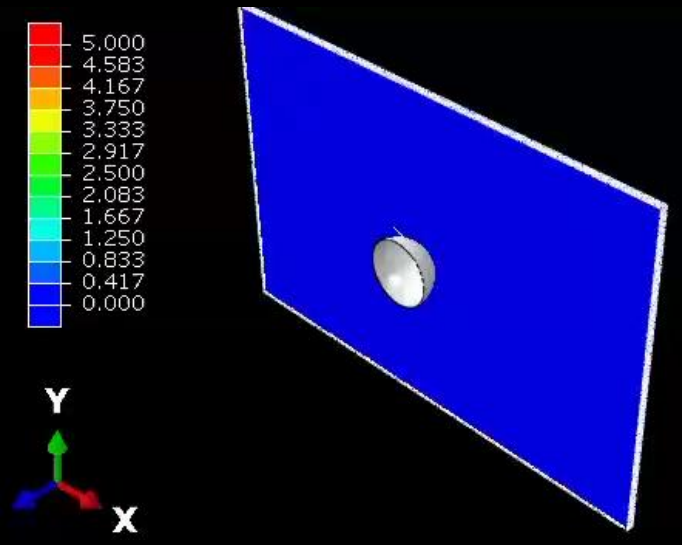
Summary & Conclusions



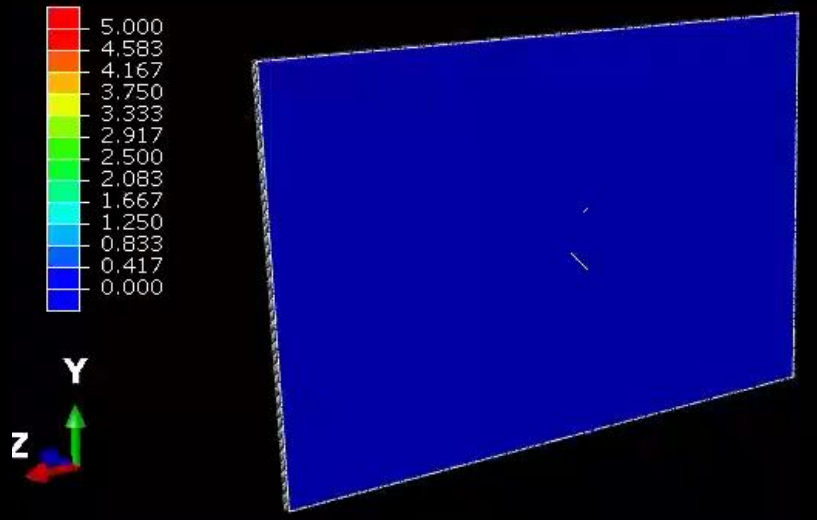
Results & Discussion







Top (impacted) surface



Bottom (back) surface

* Example result. The FEA-predicted maximum failure indices of a baseline QI specimen under 15J LVI.

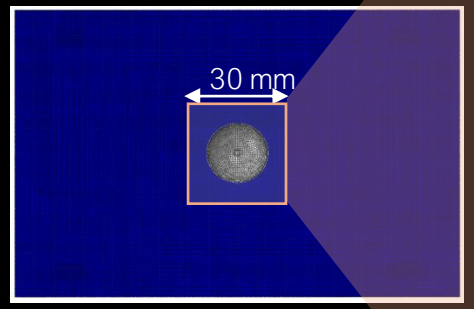
Results & Discussion

- FEA prediction of LVI (15J)

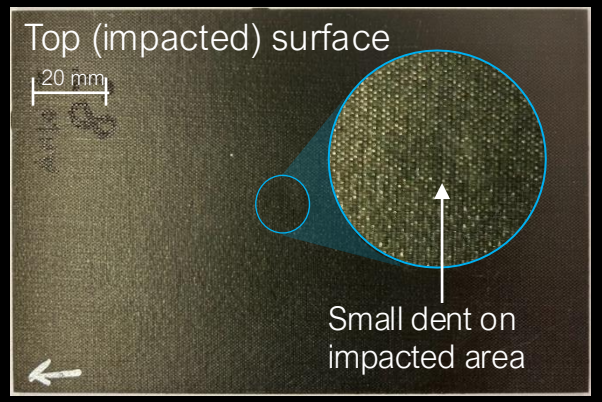
QI

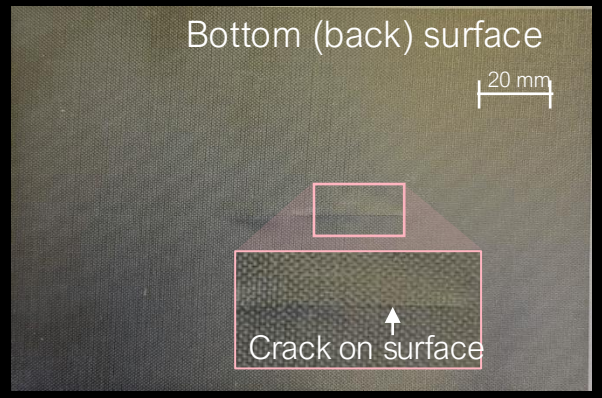
HELv1

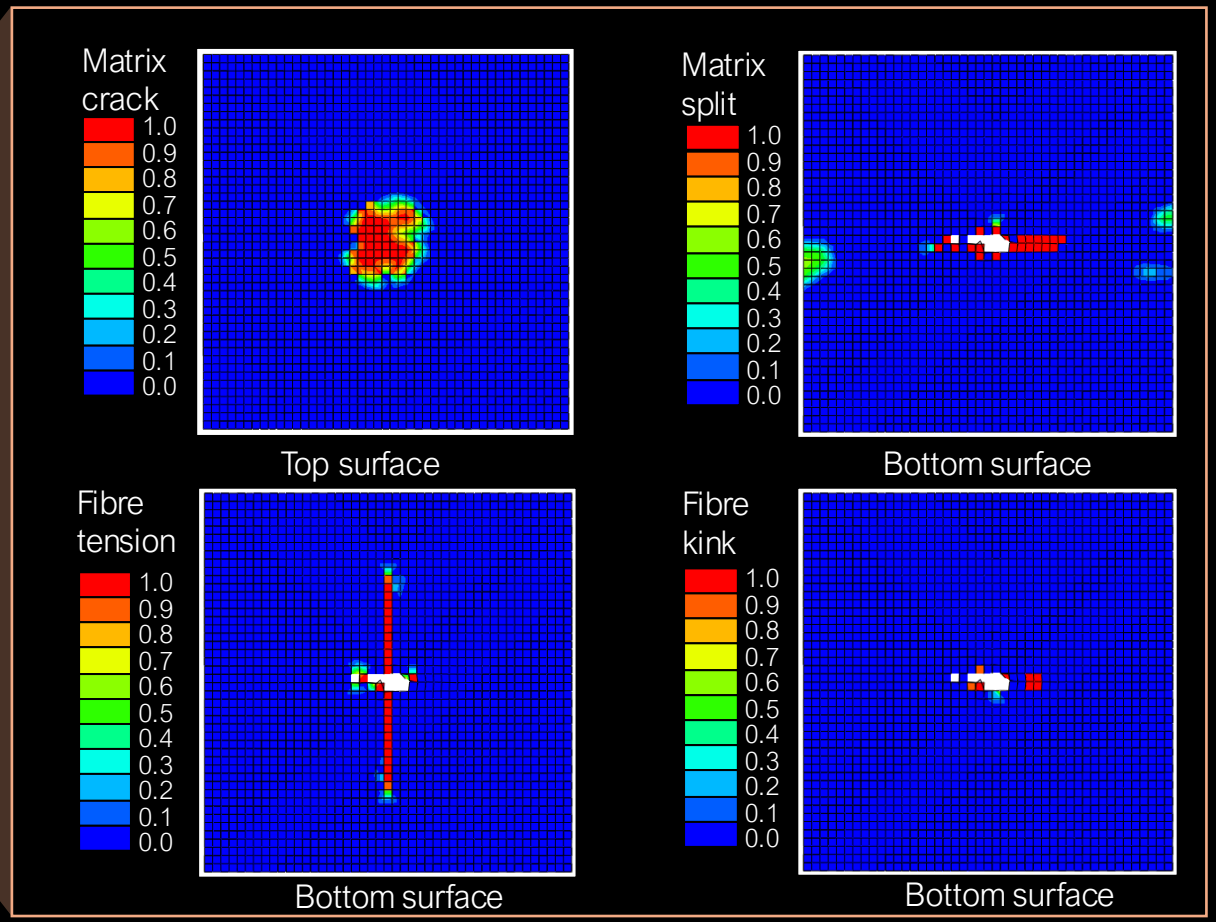
HELv2



FEA prediction







The FEA predictions capture the observed BVID pattern and dominant failure modes.

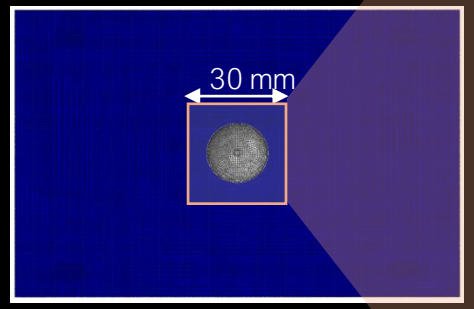
Results & Discussion

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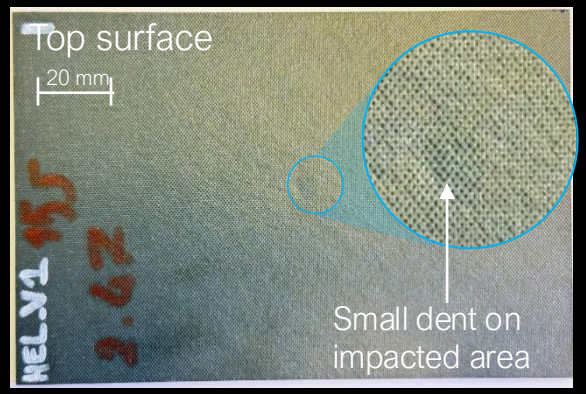
QI

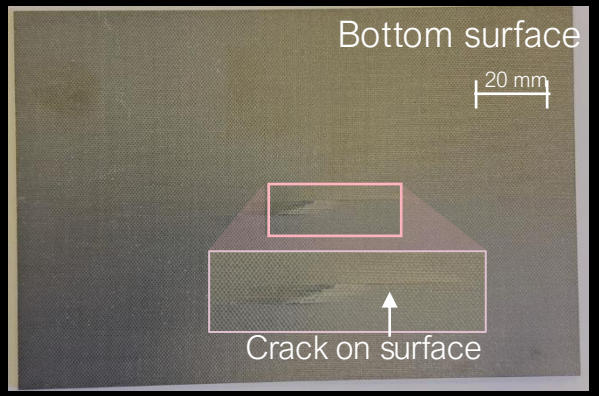
HELv1

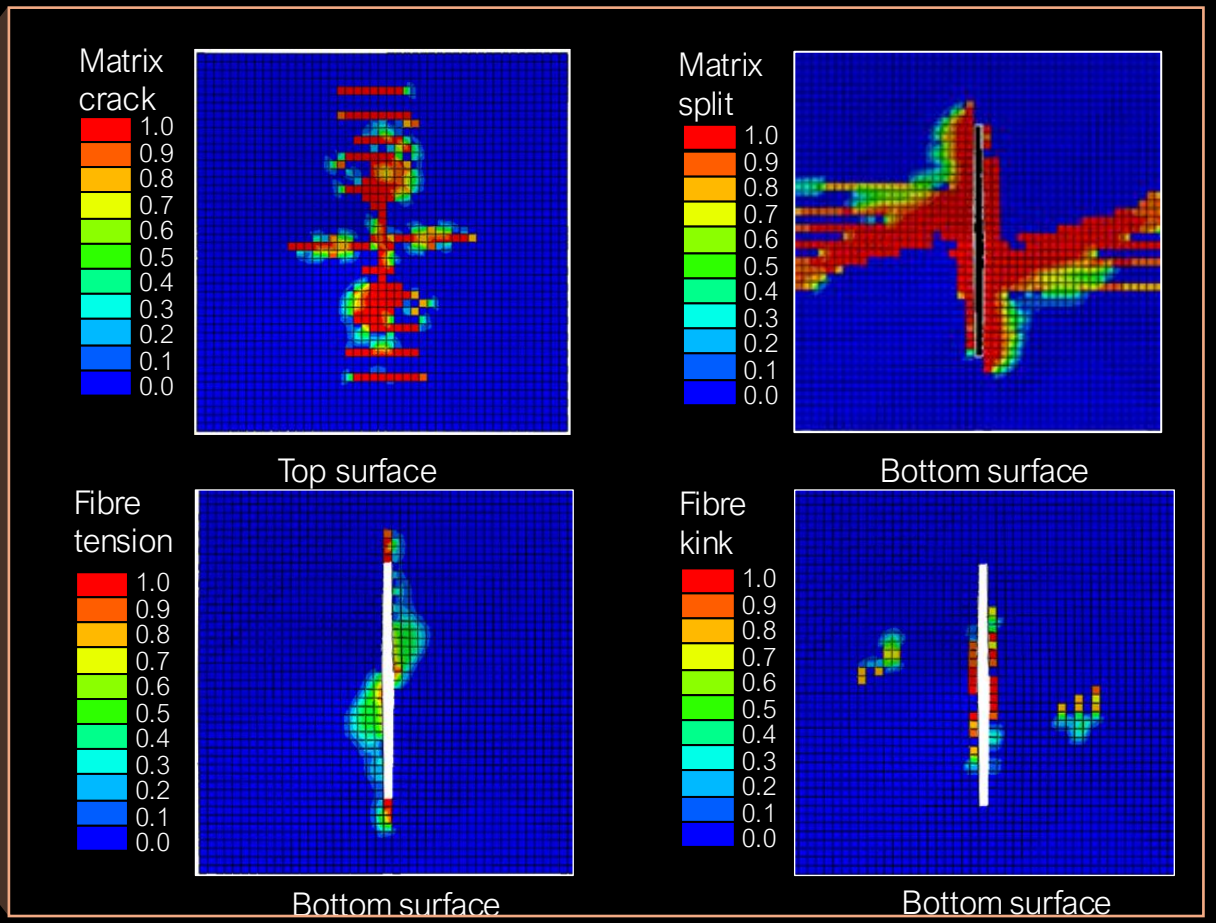
HELv2



FEA prediction







The FEA predictions capture the observed BVID pattern and dominant failure modes.

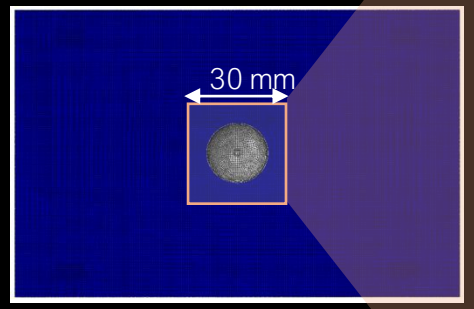
Results & Discussion

- FEA prediction of LVI (15J)

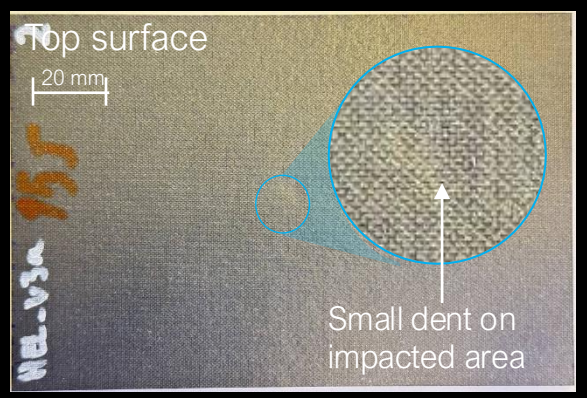
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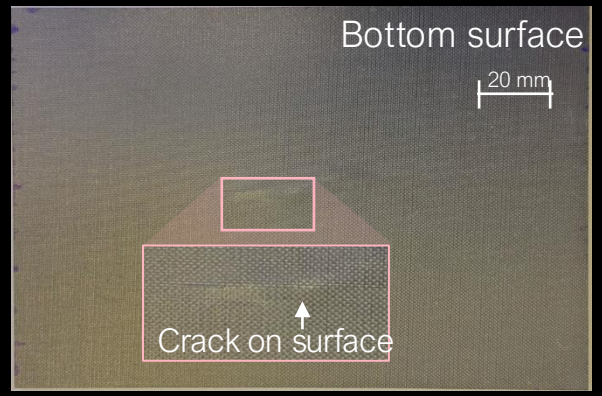
HELv1

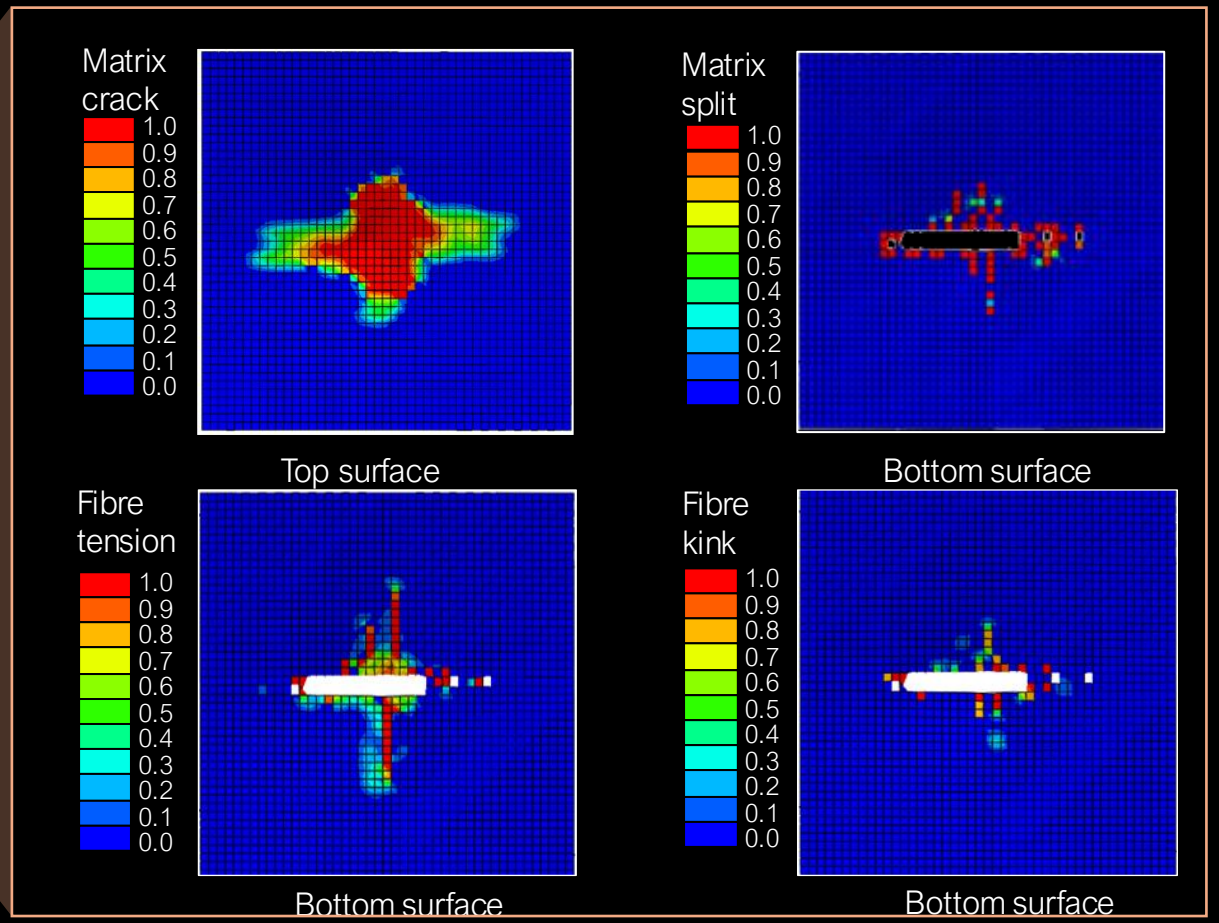
HELv2



FEA prediction



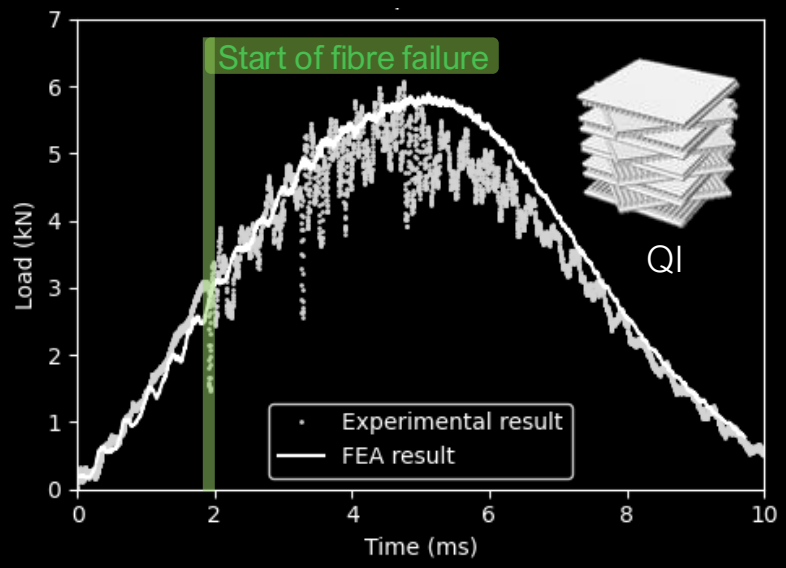


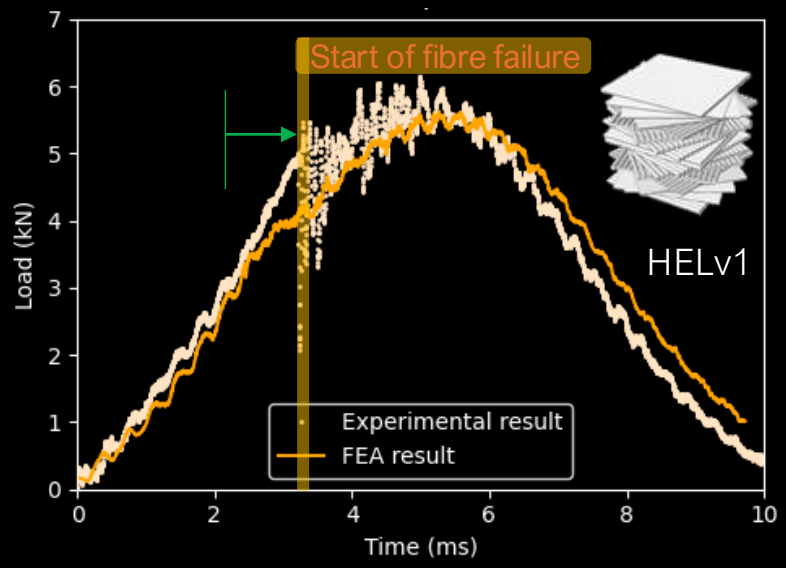


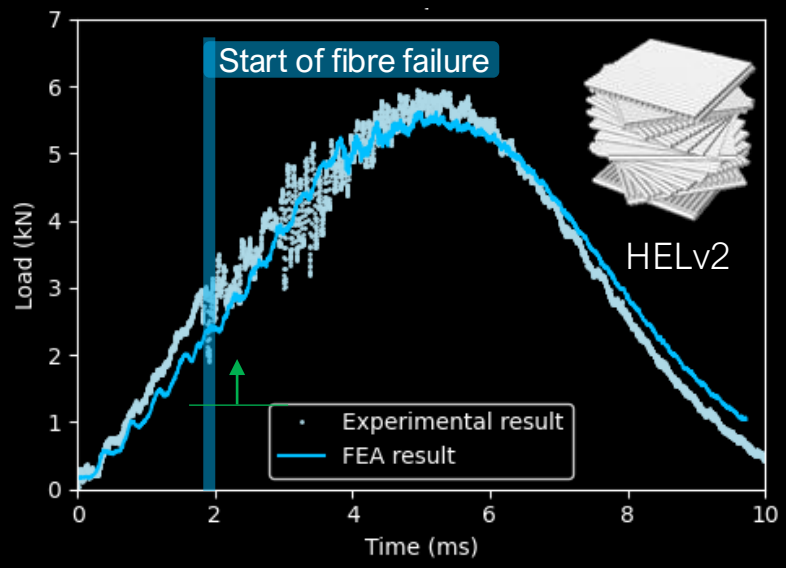
The FEA predictions capture the observed BVID pattern and dominant failure modes.

Results & Discussion

- FEA prediction of LVI (15J)







- FEA captures the overall load–time response.
- HELv1 delays fibre failure onset from ~2.0 to ~3.5 ms.
- HELv2 shows a smaller post-peak load drop.
- Both HEL layups produce smoother responses than QI.

- Load–time responses suggest different failure progressions.
- Energy dissipation, contact damage, and in-ply damage are examined next.

Results & Discussion

Energy Dissipation | Contact Damage | In-ply damage

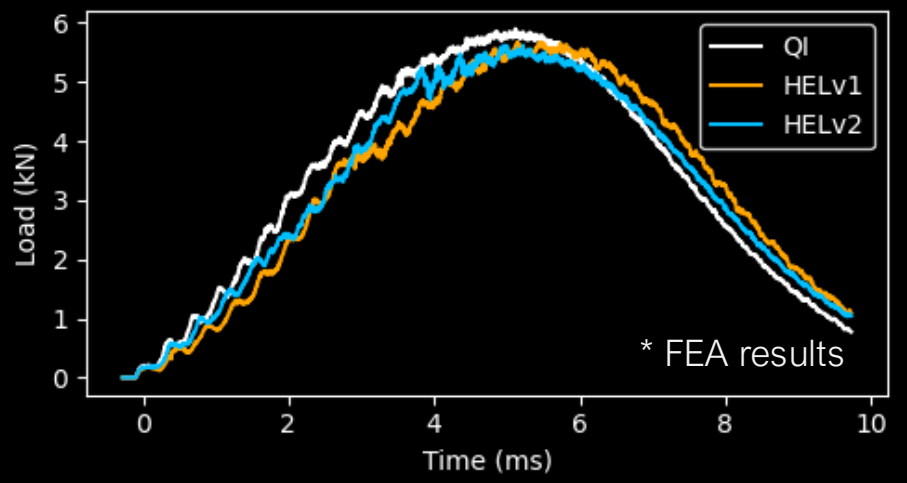
The energy curves corroborate the experimental observations

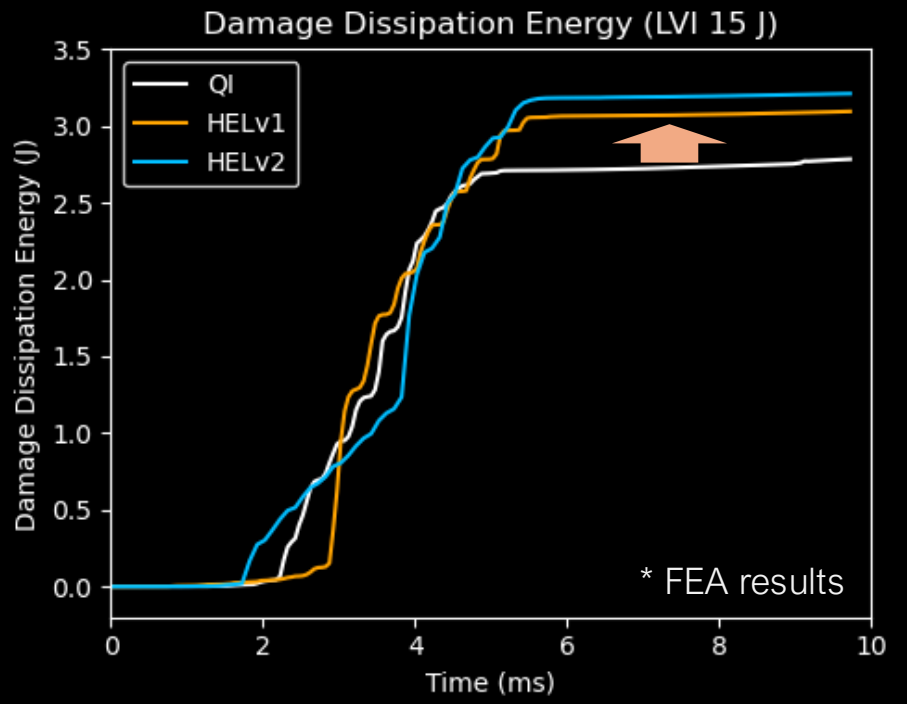
- HELv1 delays the onset of damage
- HELv2 shows smoother dissipation process

Both HEL layups dissipate more energy, indicating that a smaller pitch angle spreads localised damage over a wider area

Hypothesis [1]

More energy dissipation ← More damage distribution → May enhance post-impact performance



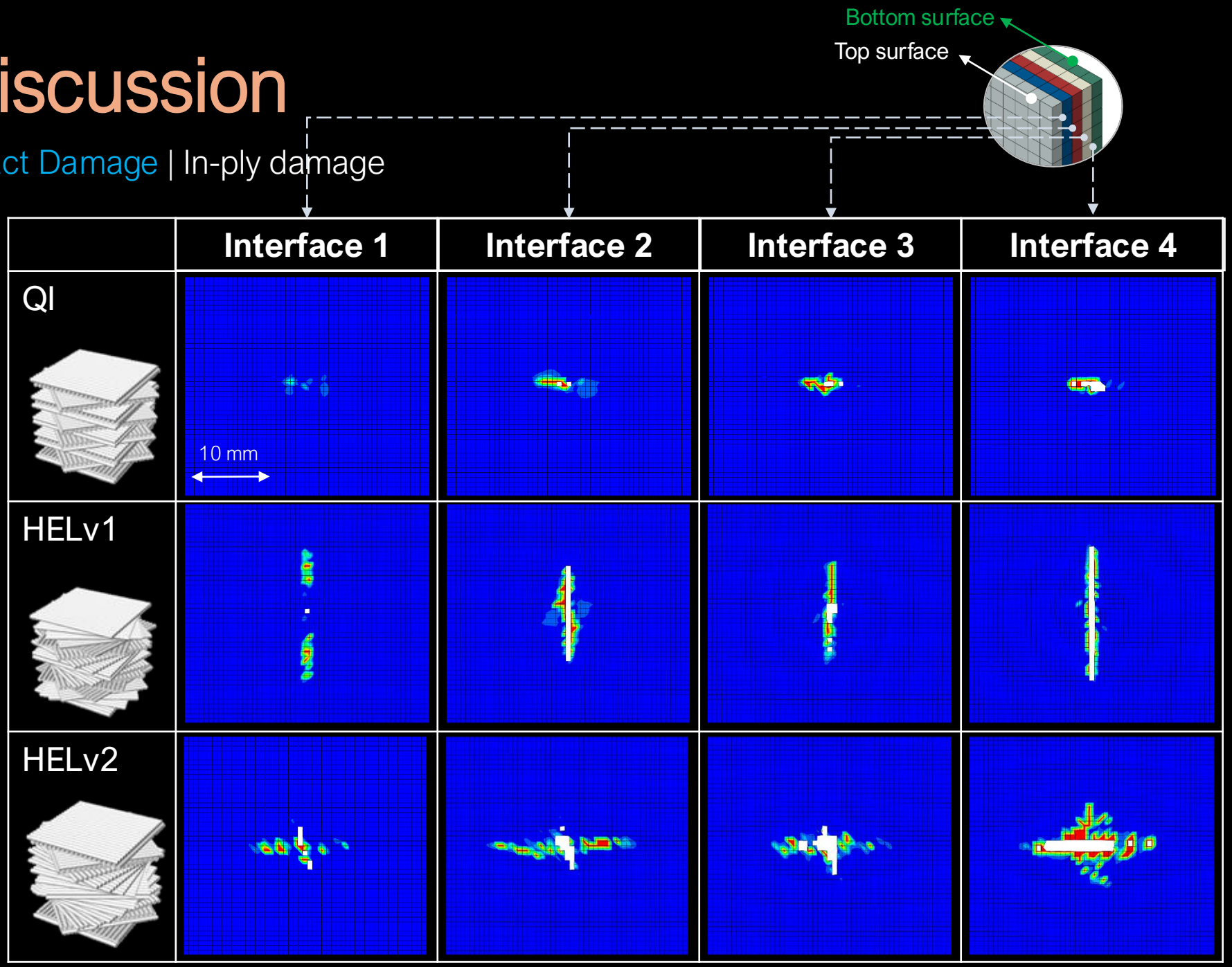


[1] Mencattelli, L. and S.T. Pinho, Herringbone-Bouligand CFRP structures: A new tailorable damage-tolerant solution for damage containment and reduced delaminations. Composites Science and Technology, 2020. 190: p. 108047.

Results & Discussion

Energy Dissipation | Contact Damage | In-ply damage

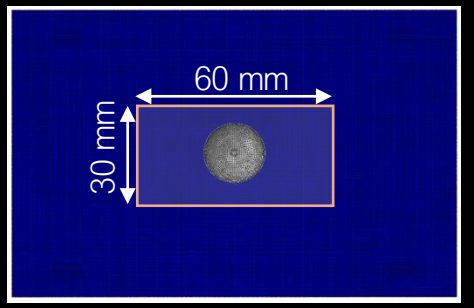
- HEL layups show wider inter-ply damage
- Damage spreads across multiple interfaces



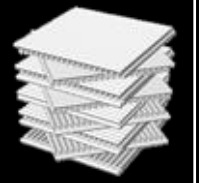
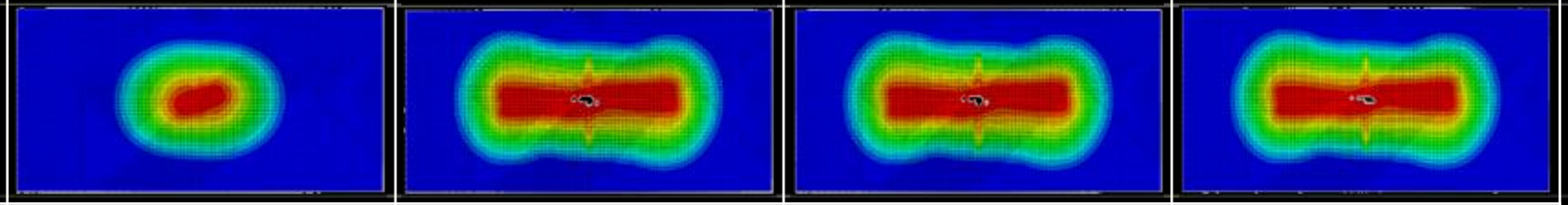
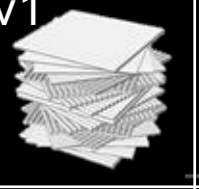
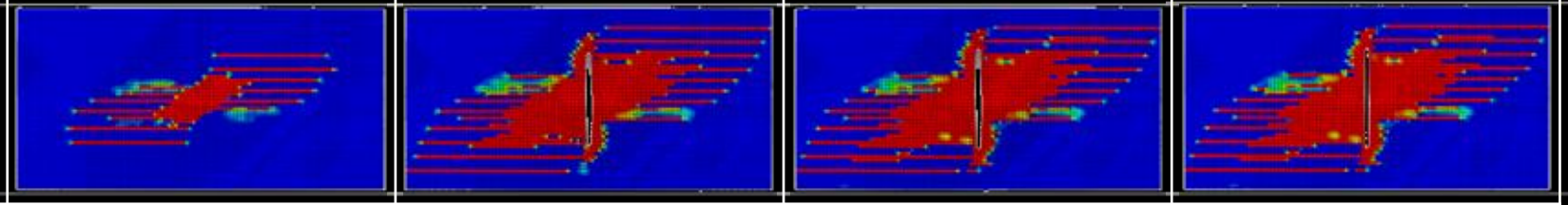
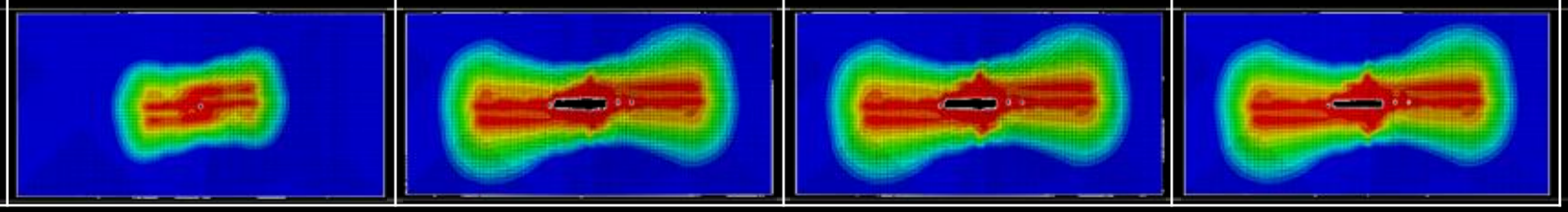
Results & Discussion

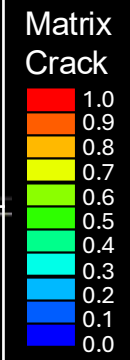
Energy Dissipation | Contact Damage | In-ply damage

- Different layups produce distinct matrix-cracking patterns.
- Compared with QI, HELv1 and HELv2 show larger matrix-cracking regions.
- The helicoidal layups redistribute matrix cracking over a wider area.



Matrix-cracking damage history at four time points for the three layups on the bottom surface.

	2.5 ms	5 ms	7.5 ms	10 ms
QI				
HELv1				
HELv2				

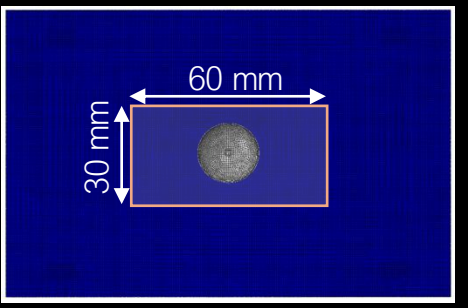


Results & Discussion

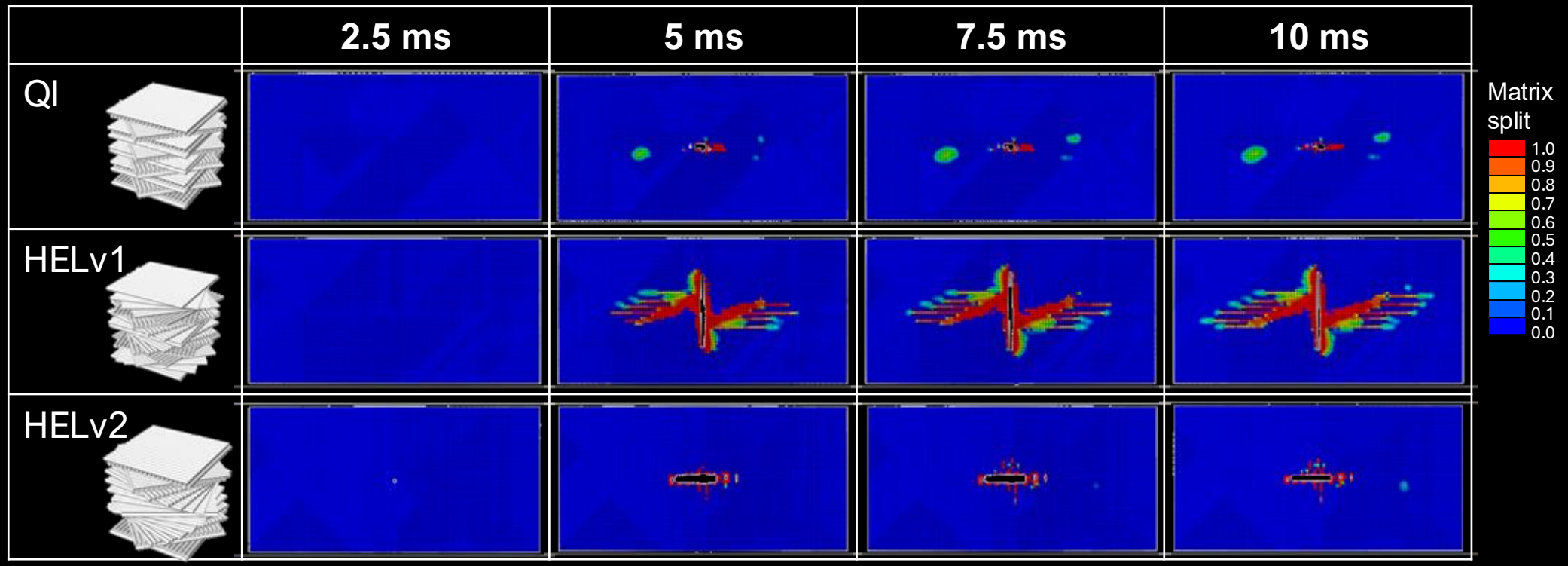
Energy Dissipation | Contact Damage | In-ply damage

- Compared with QI, HELv1 and HELv2 show more distributed matrix-splitting damage.

More energy dissipation <----- More damage distribution

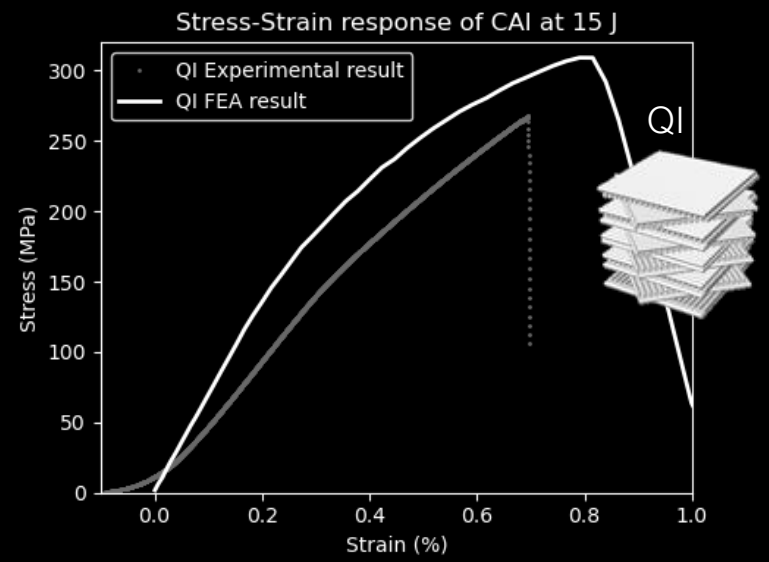


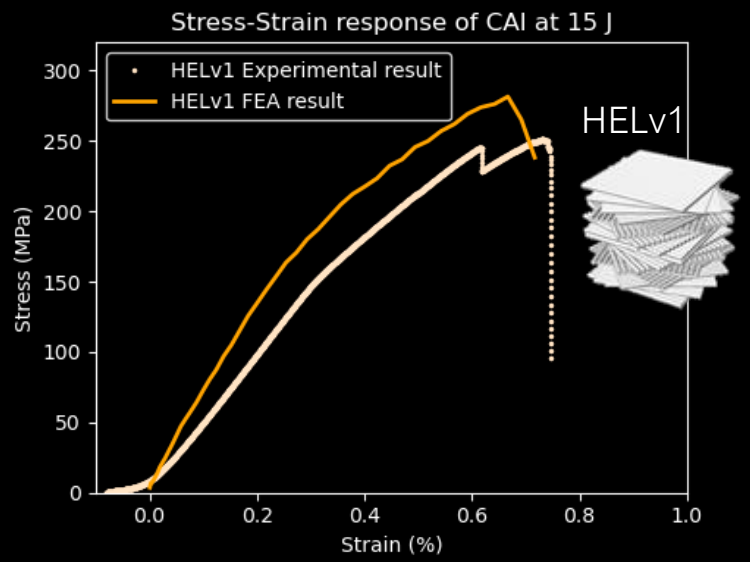
Matrix-splitting damage history at four time points for the three layups on the bottom surface.

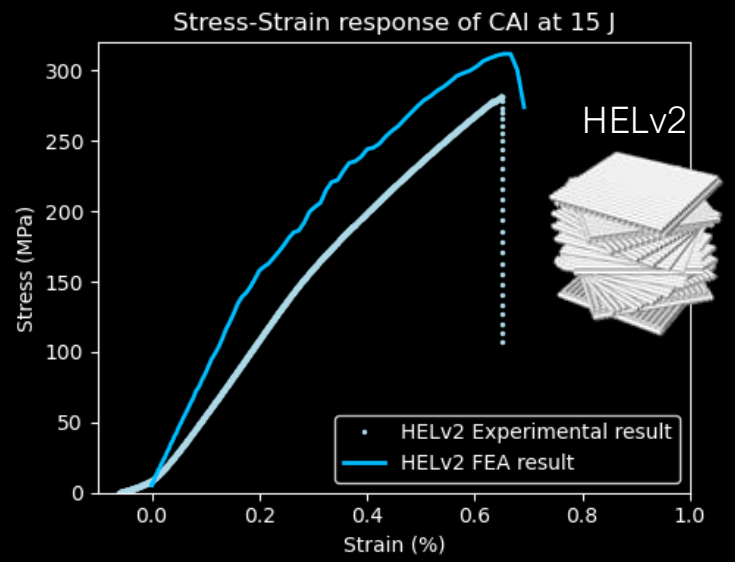


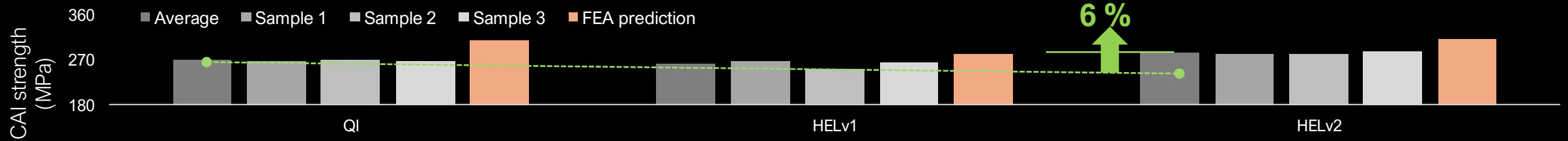
Results & Discussion

- FEA prediction of CAI at 15J
  - The CAI stress–strain curves from FEA are compared to the experimental results
  - FEA reproduces the overall CAI response and layup strength ranking
  - Small stiffness differences may reflect test-machine and fixture compliance
  - HELv2 achieves a 6% CAI-strength increase over QI









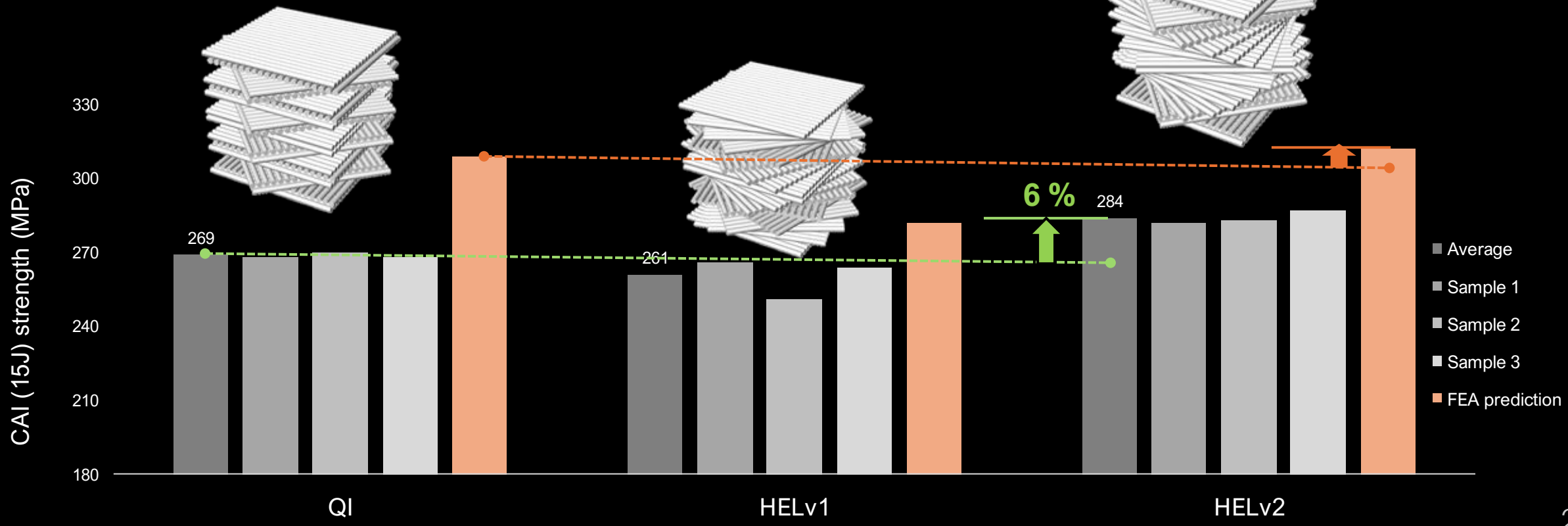
Results & Discussion

- FEA prediction of CAI at 15J
  - FEA reproduces the overall CAI response and layup strength ranking
  - Small stiffness differences may reflect test-machine and fixture compliance
  - HELv2 achieves a 6% CAI-strength increase over QI

More damage distribution → May enhance post-impact performance



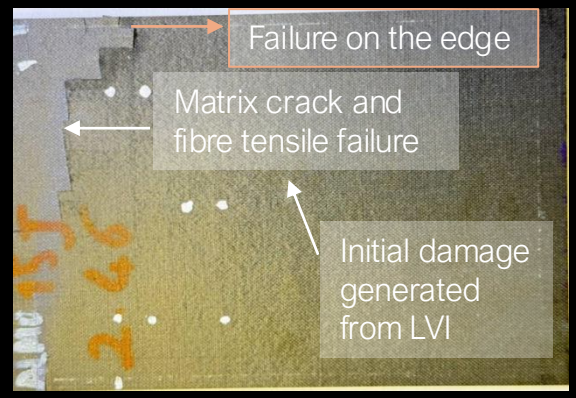


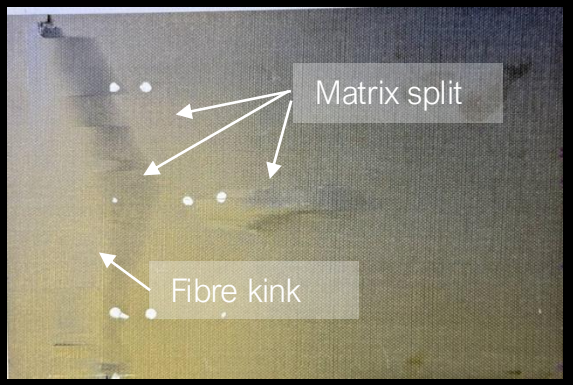


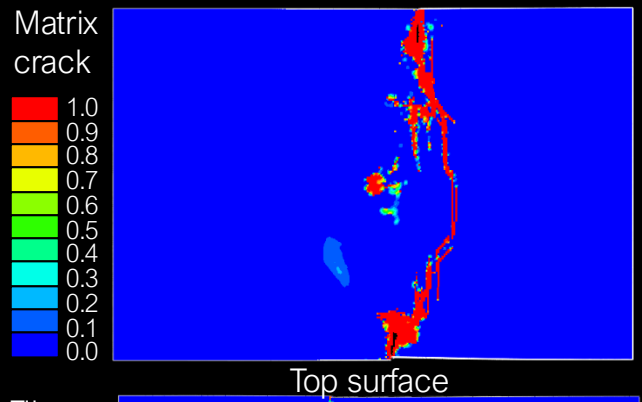
Results & Discussion

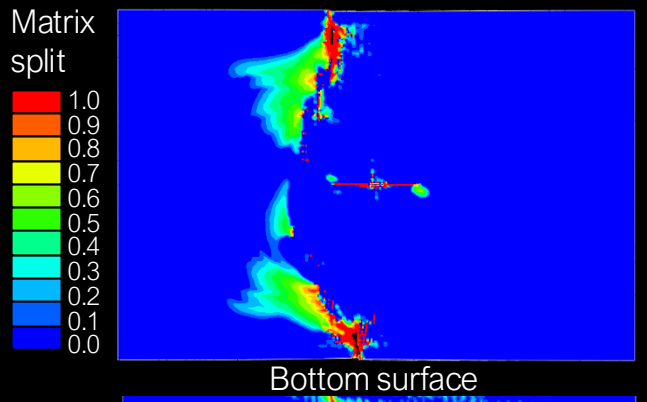
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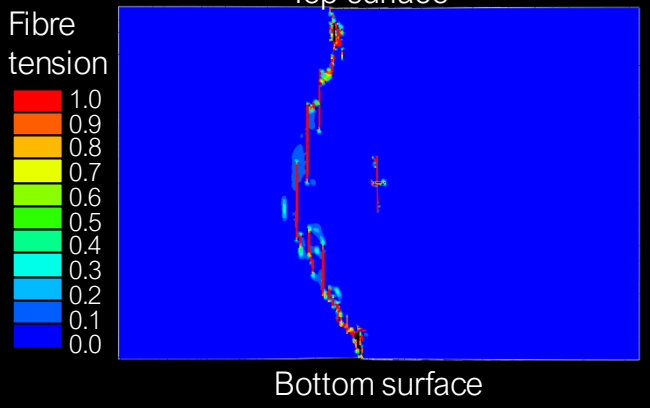
- Post-test inspection suggests that QI failure initiates near the clamped region.
- Matrix cracking, matrix splitting, fibre tension and fibre kinking are observed consistently across all tested specimens.
- The right-hand figures show the FEA-predicted damage variables for the QI layup at final FEA failure state, by failure mode

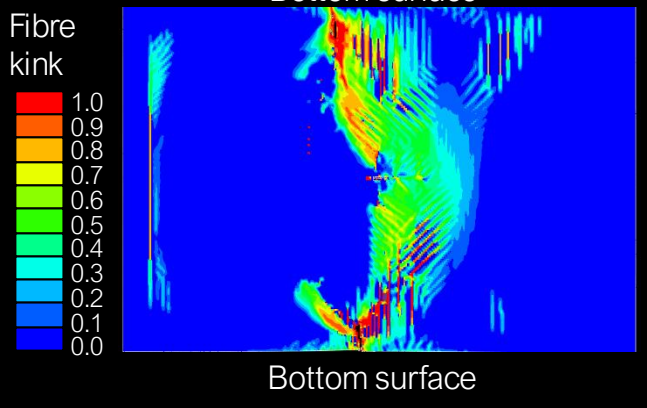




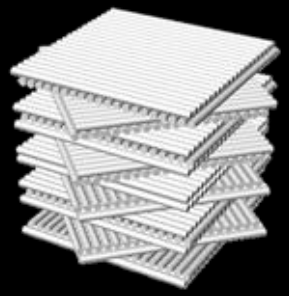








QI

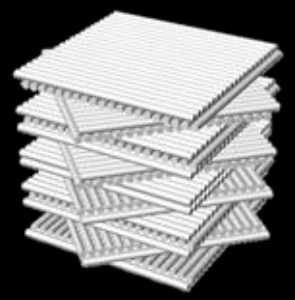


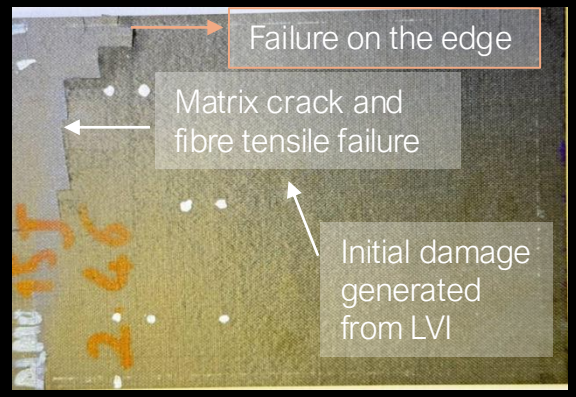
Results & Discussion

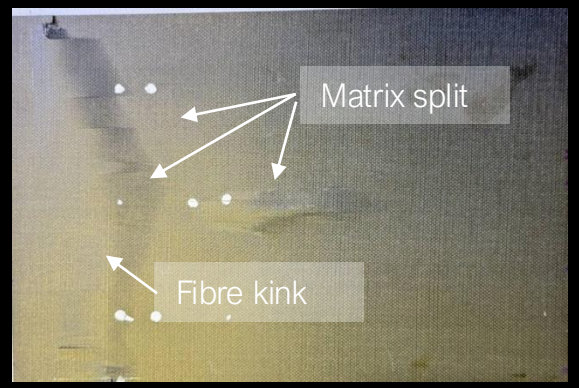
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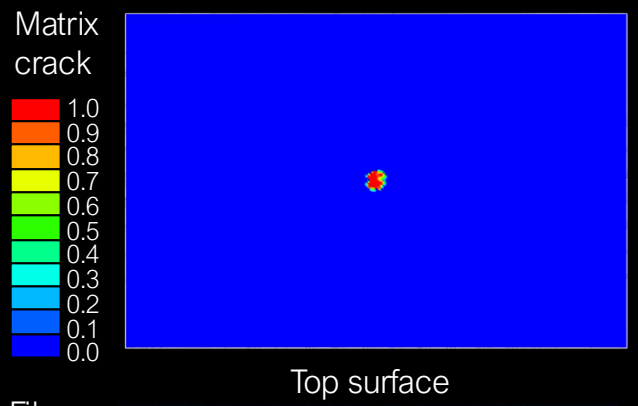
- Post-test inspection suggests that QI failure initiates near the clamped region.
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- The animations show the FEA-predicted evolution of damage variables for the QI layup, by failure mode.

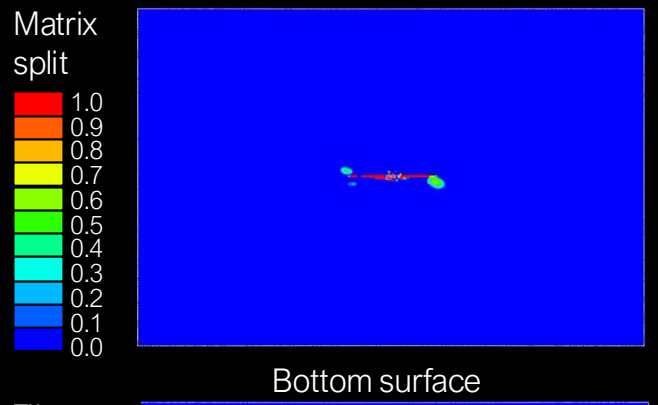
QI

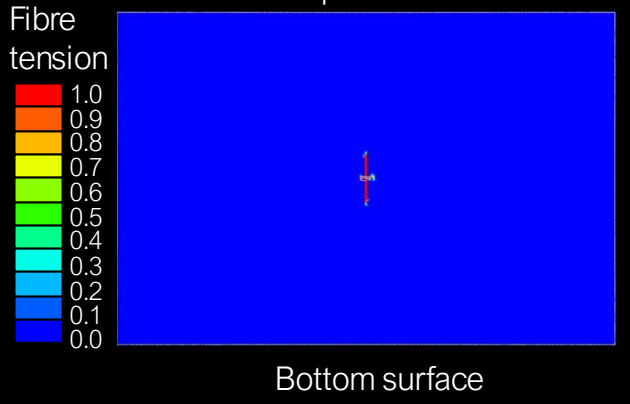


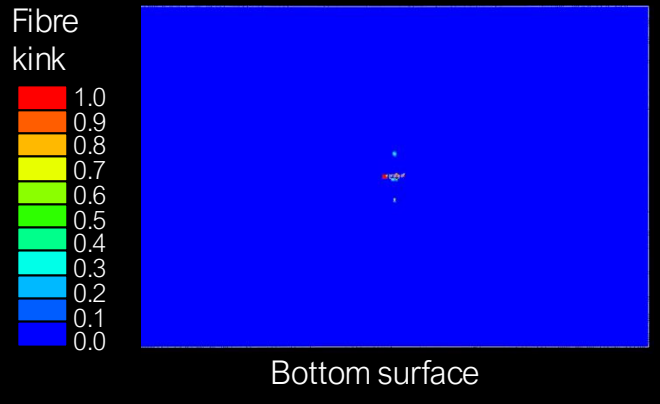






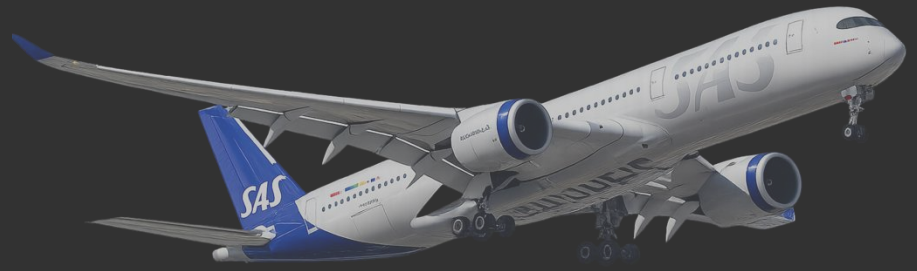




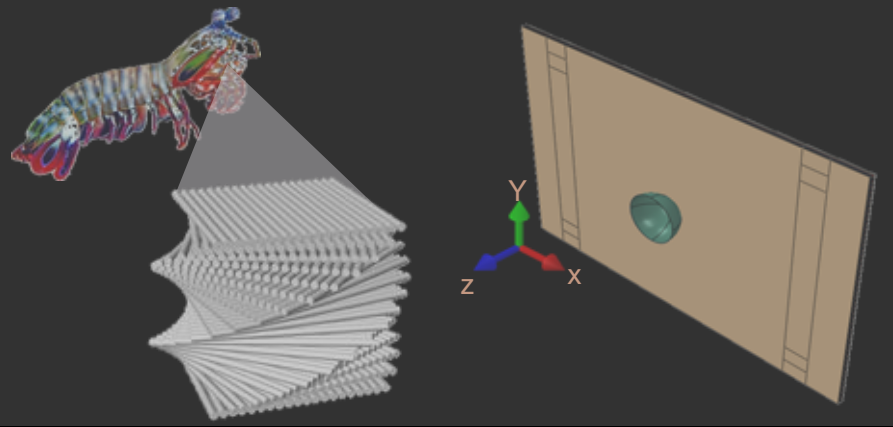


Outline

Background & Motivation



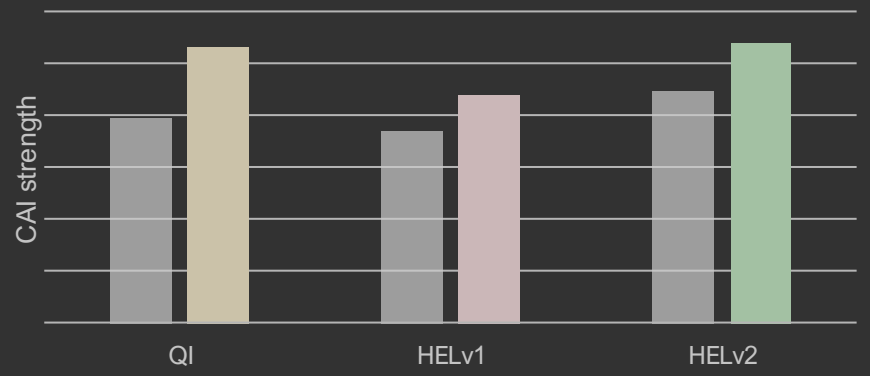
Design & Methodology



Results & Discussion



Summary & Conclusions

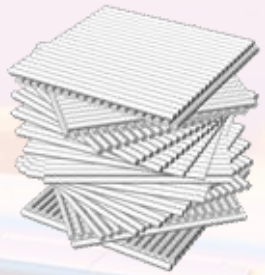


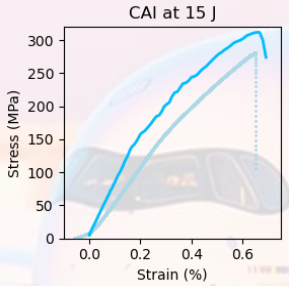
Design and FEA of Bio-inspired Carbon Fibre-Reinforced Polymer (CFRP) Laminates for Improved Low-Velocity Impact (LVI) Damage Tolerance

IMPERIAL

ECCM22
22nd EUROPEAN CONFERENCE ON COMPOSITE MATERIALS
24th June, Oslo, Norway

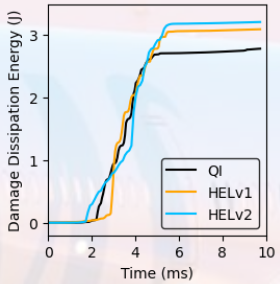
Summary

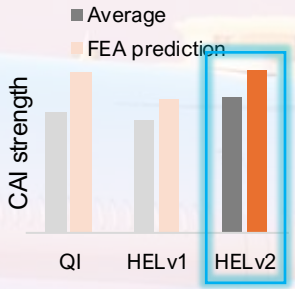




- Designed the QI and helicoidal CFRP layups;
- Conducted LVI and CAI experiments;
- Developed an FEA framework and correlated its predictions with the experimental results

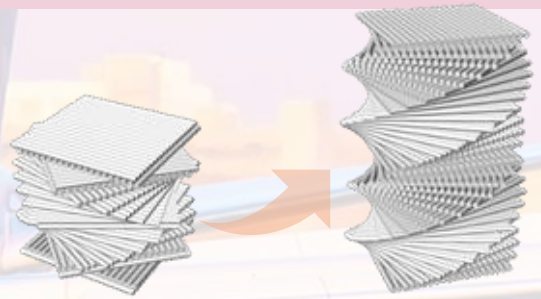
Conclusions





- HELv2 showed improved LVI damage tolerance compared with the QI baseline.
- HELv2 achieved a 6% increase in residual CAI strength.
- A smaller pitch angle redistributes impact damage over a larger area, reducing the severity of stress-concentration.

Upcoming activities



- Design thicker laminates for aerospace applications to enable a larger helicoidal design space.
- Run FEA simulations using the developed framework.
- Apply the helicoidal concept to a wing sub-element demonstrator to assess structural-level scalability.

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@hongyan-wang18
https://github.com/Hongyan-Wang
https://pinholab.cc.ic.ac.uk

UKRI-Innovate project - IVI
(Isothermal Variable-volume Infusion)

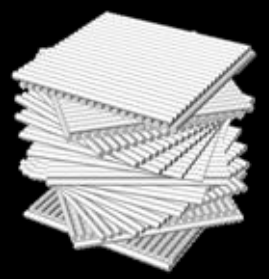
HELICOID INDUSTRIES LTD
Short Brothers A Boeing Company
Department for Business & Trade
AEROSPACE TECHNOLOGY INSTITUTE
UKRI Innovate UK

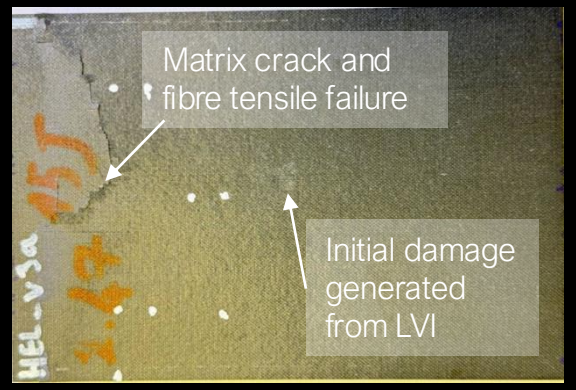
Results & Discussion

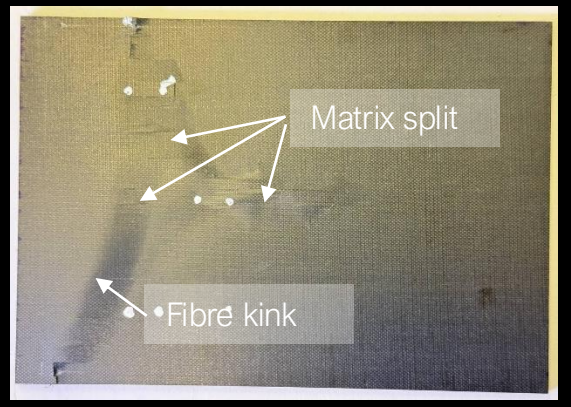
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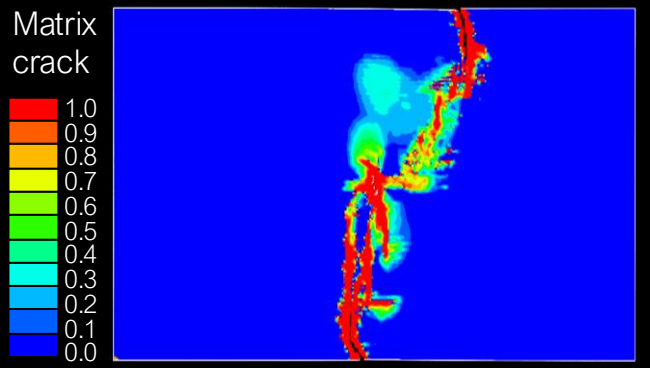
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- Matrix cracking, matrix splitting, fibre tension and fibre kinking are observed consistently across all tested specimens.
- The right-hand figures show the FEA-predicted damage variables for the HELv2 layup at final FEA failure state, by failure mode

HELv2

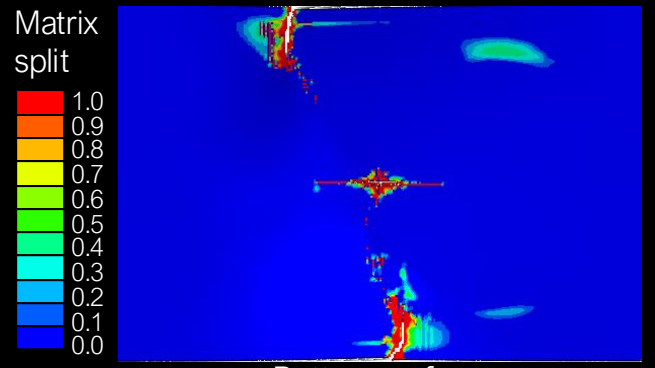




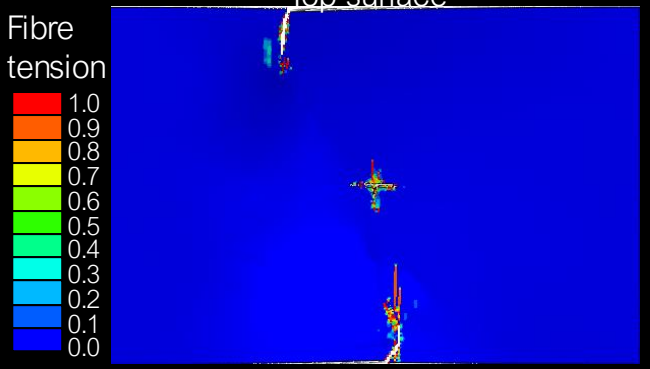




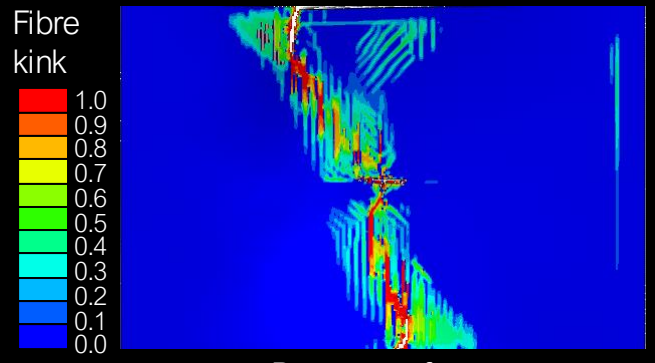
Top surface



Bottom surface



Bottom surface



Bottom surface

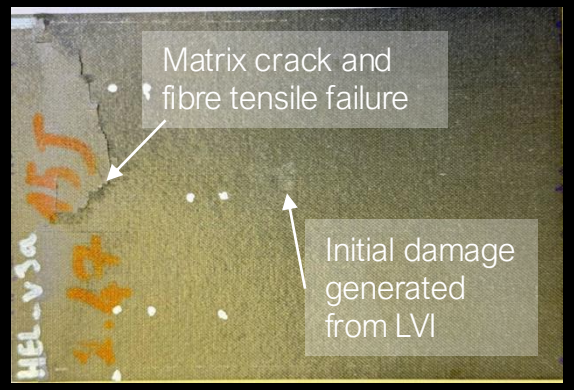
Results & Discussion

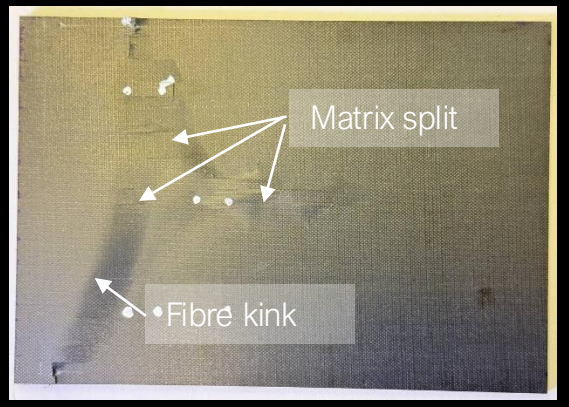
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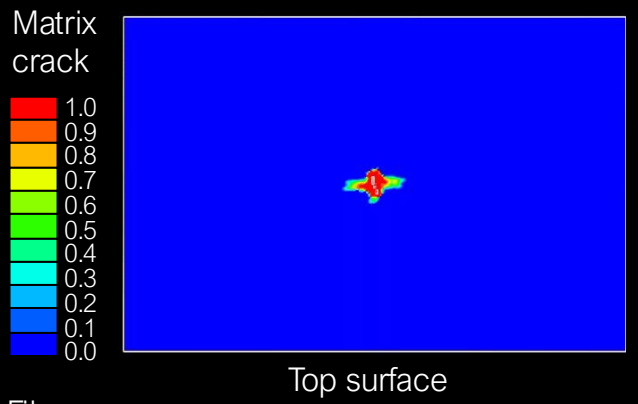
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HELv2

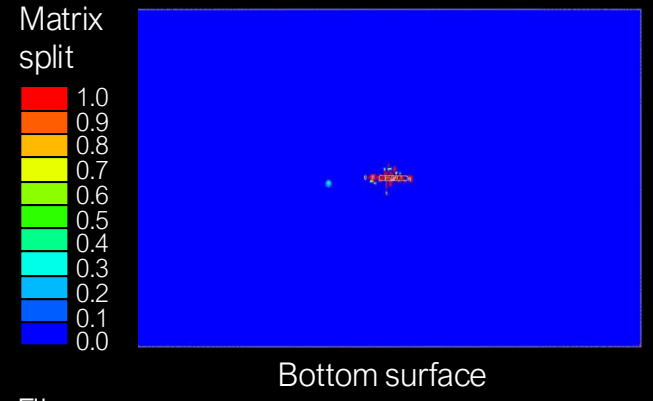




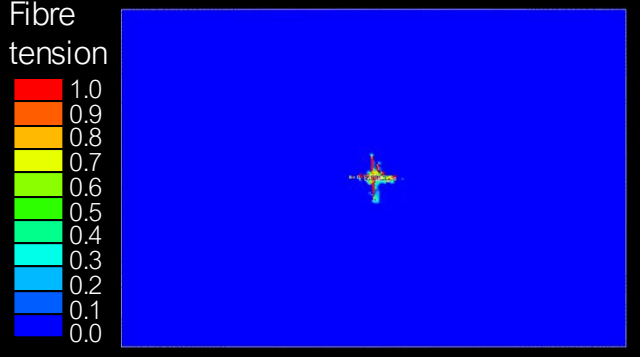




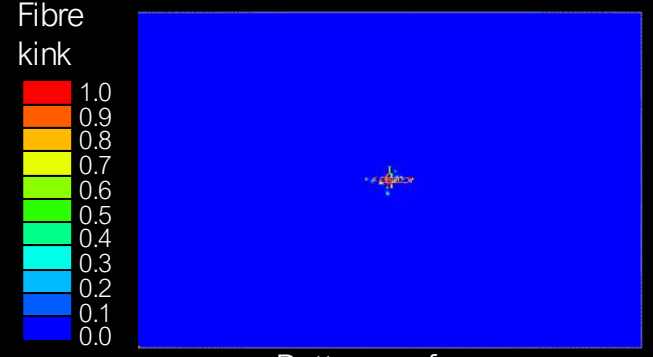
Top surface



Bottom surface

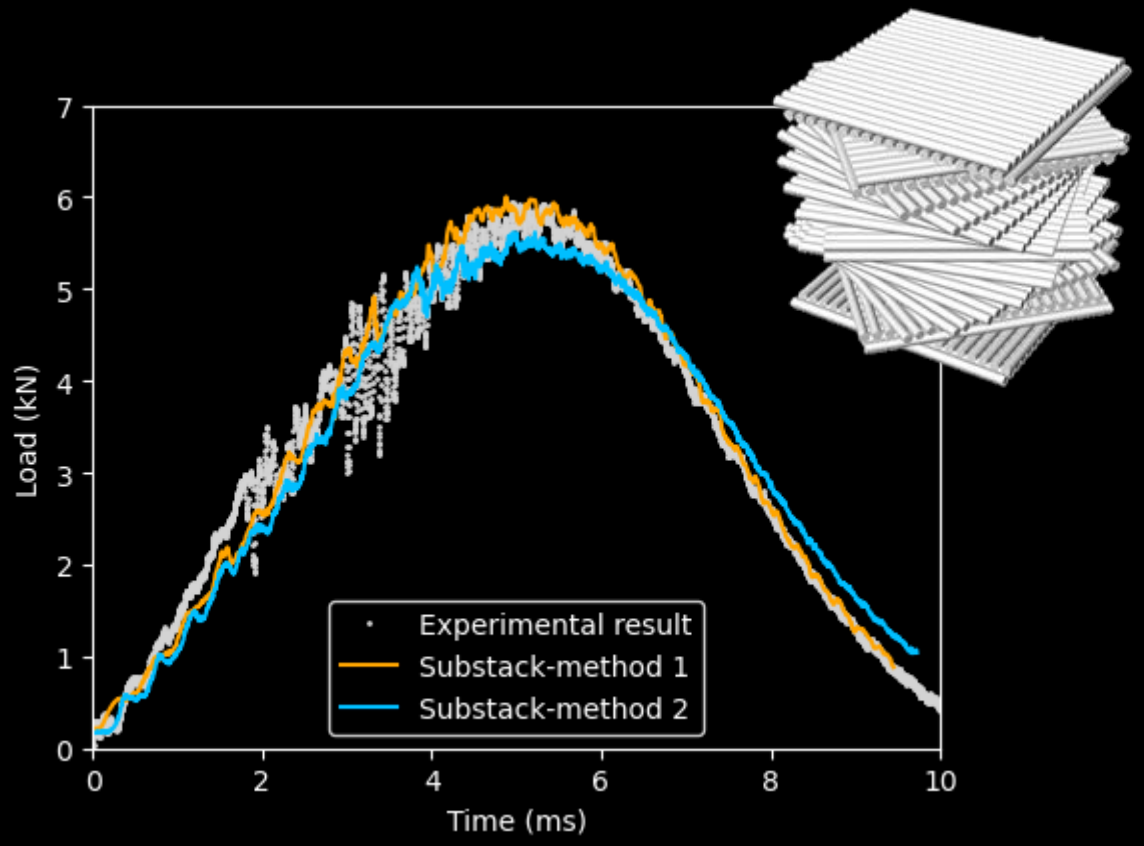


Bottom surface

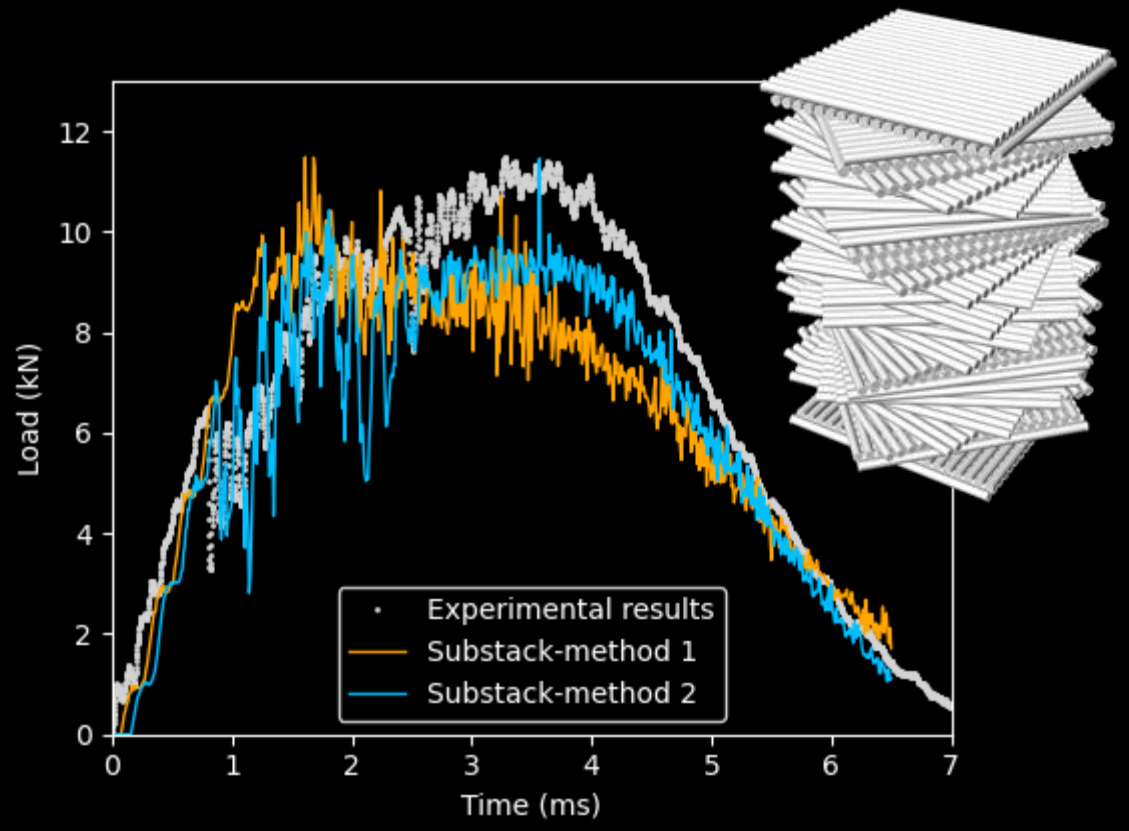


Bottom surface

Appendix 1 – substack study



20-layer laminate



33-layer laminate

- Substack-method 1 composes of 5 layers per substack
- Substack-method 2 composes of 3-4 layers per substack ✓